

SGA 3, Exposé XI
Representability Criteria
Applications to Multiplicative-Type Subgroups
of Affine Group Schemes
English Reconstruction

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Exposé XI

Representability Criteria

Applications to

Multiplicative-Type Subgroups

of Affine Group Schemes

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0. Introduction

As we have already seen in Exposé X, nos. 4 and 5, the representability of certain functors—for example, functors of the form $\mathrm{Hom}_S(X, Y)$ and various related constructions—plays an important role in many questions concerning group preschemes.

Among the results that are particularly useful in this theory, besides the representability of quotients studied in Exposés V and VI_B and in Exposé VIII, no. 5, let us mention the representability of functors of the form

$$\prod_{X/S} Y/X,$$

where Y is a subobject of X . This was studied in a very elementary case in Exposé VIII, no. 6; variants will be given in no. 6 of the present exposé. These results provide the representability of various centralizers, normalizers, and transporters.

Less elementary representability criteria, using results that will appear in EGA VI, are indicated in 6.12 and in Exposés XV and XVI. There we shall give a criterion for the representability of quotients G/H in cases not covered by the preceding exposés; this criterion was not developed in the oral lectures.

Our principal purpose in the present exposé is to prove Theorems 4.1 and 4.2, which provide a typical example of a nonprojective construction technique, close to the technique that will be developed in EGA VI. Since the oral lecture and the drafting of the present text, it has moreover become apparent that the affine hypotheses in 4.1 and 4.2 can be removed to a large extent (cf. Exposé XV), and that the main part of the theory developed in the following exposé can dispense with 4.1 and 4.2. Finally, it might be interesting to prove analogues of these results for a general reductive group prescheme (for example, a semisimple one) in place of a group of multiplicative type; in that case 4.1 and 4.2 would doubtless be the key result in the proof.

⁰Editor's note: xy version of December 5, 2008.

1. Review of Smooth, Étale, and Unramified Morphisms

The reader is referred to EGA IV, §§17 and 18, and, pending its publication, to SGA 1, Exposés I, II, and III. Certain noetherian hypotheses used there, which are inconvenient in applications, should however be replaced by finite-presentation hypotheses.

Definition 1.1. — *Let S be a prescheme and let*

$$F : (\mathbf{Sch})^\circ / S \longrightarrow (\mathbf{Ens})$$

be a functor. The functor F is said to be formally smooth (respectively, formally unramified¹, respectively, formally étale) if, for every S -prescheme S' that is affine in the absolute sense and every subscheme $S'_0 \subset S'$ defined by a nilpotent ideal $\mathcal{J}$, the map

$$F(S') \longrightarrow F(S'_0)$$

is surjective (respectively, injective, respectively, bijective).

An S -prescheme X is said to be formally smooth over S (respectively, formally unramified over S , respectively, formally étale over S) if the corresponding functor has that property. It is said to be smooth over S (respectively, unramified over S , respectively, étale over S) if it satisfies the preceding condition and is, in addition, locally of finite presentation over S .

The usefulness of these definitions for X lies, on the one hand, in their remarkably simple expression in terms of the functor represented by X (and in practice X is often given as the object over S representing an explicitly described functor), and, on the other hand, in the remarkable local structural properties that they express. We recall those properties in the following statements; for their proofs, see the references just cited.

Proposition 1.2. — *Let X be a prescheme locally of finite presentation over S . Then:*

- (i) *X is smooth over S if and only if it is flat over S and its geometric fibers*

$$X \otimes_S \mathrm{Spec} \kappa(s)$$

are regular schemes. More generally, X is smooth over S in a neighborhood of $x \in X$ (one then says that X is smooth over S at x) if and only if X is flat over S at x and X_s is smooth over $\kappa(s)$ at x ; that is,

$$X_s \otimes_{\kappa(s)} \overline{\kappa(s)}$$

is regular at the points (or merely at one point) lying over x .

- (ii) *Suppose that S , and hence X , is locally noetherian. Let $x \in X$, and let $s \in S$ be its image. The smoothness of X over S at x can then be detected from the local homomorphism*

$$A = \mathcal{O}_{S,s} \longrightarrow B = \mathcal{O}_{X,x}$$

of noetherian local rings (or even from the local homomorphism $\hat{A} \rightarrow \hat{B}$ of their completions), by the following characteristic property: B is flat over A , and $B \otimes_A k$ is geometrically regular over k , the residue field of A . Equivalently, for every finite extension k' of k ,

$$(B \otimes_A k) \otimes_k k' = B \otimes_A k'$$

¹The source notes that one would now rather say “net” in place of “unramified.”

is a regular semilocal ring. When the residue-field extension $k(B)/k(A)$ is trivial, these conditions are also equivalent to the following one: $\widehat{B}$ is isomorphic, as an $\widehat{A}$ -algebra, to an algebra of formal power series

$$\widehat{A}[[t_1, \dots, t_n]].$$

Thus, from the “formal” point of view, the structure of X over S is that of the standard affine space $S[t_1, \dots, t_n]$ over S .

Proposition 1.3. — *Let X be a prescheme locally of finite presentation over S . Then:*

- (i) *The following conditions are equivalent:*
 - (a) *X is unramified over S .*
 - (b) *The diagonal morphism $X \rightarrow X \times_S X$ is an open immersion.*
 - (c) $\Omega_{X/S}^1 = 0$.
 - (d) *The geometric fibers of X/S are discrete and reduced; that is, they are isomorphic to disjoint sums of copies of the base field.*
 - (e) *For every $s \in S$, the fiber*

$$X \otimes_S \text{Spec } \kappa(s) = X_s$$

is unramified over $\kappa(s)$, or equivalently is isomorphic to a disjoint sum of spectra of finite separable extensions of $\kappa(s)$.

- (ii) *There are analogous pointwise conditions for X to be unramified over S at a given point x , that is, in a neighborhood of that point. For example, it is necessary and sufficient that $\mathcal{O}_{X_s, x}$ be a finite separable extension of $\kappa(s)$. In terms of the local homomorphism*

$$A = \mathcal{O}_{S, s} \longrightarrow B = \mathcal{O}_{X, x},$$

this says that $B \otimes_A k$ is a finite separable extension of the residue field k of A ; equivalently,

$$\mathfrak{r}(A)B = \mathfrak{r}(B) \quad \text{and} \quad k(B)/k(A) \text{ is finite separable.}$$

When A , and hence B , is noetherian and $k(A) \xrightarrow{\sim} k(B)$, this also means that $\widehat{A} \rightarrow \widehat{B}$ is surjective.

Thus, from the “formal” point of view, saying that X is unramified over S means that X is essentially a subprescheme of S . Observe also that conditions (d) and (e) are expressed solely in terms of the fibers of X/S .

Proposition 1.4. — *Let X be a prescheme locally of finite presentation over S . Then X is étale over S if and only if it is smooth and unramified over S (which is immediate from the definition), so the criteria in 1.2 and 1.3 apply. In particular:*

- (i) *X is étale over S if and only if it is flat and unramified over S . There is an analogous local criterion for X to be étale over S at a given point x .*

- (ii) Suppose in addition that S , and hence X , is locally noetherian. Whether X is étale over S at x can then be detected from the local homomorphism

$$A = \mathcal{O}_{S,s} \longrightarrow B = \mathcal{O}_{X,x}$$

² (and even from $\hat{A} \rightarrow \hat{B}$) by the following characteristic property: B is flat over A , and $B \otimes_A k$ is a finite separable extension of k , the residue field³ of A . When $k(A) \xrightarrow{\sim} k(B)$, this condition simply says that $\hat{A} \rightarrow \hat{B}$ is an isomorphism.

Thus, from the “formal” point of view, saying that X is étale over S simply means that X is locally isomorphic to S .

Remark 1.5. — When X is locally of finite presentation over S and a point $x \in X$ is given, the fact that X is smooth (respectively, unramified, respectively, étale) over S at x , that is, in a neighborhood of x , can also be detected from the functor

$$X : (\mathbf{Sch})^\circ / S \longrightarrow (\mathbf{Ens})$$

by the following property. For every S -scheme S' that is the spectrum of a local ring, every subscheme $S'_0 \subset S'$ defined by a nilpotent ideal, and every S -morphism $u_0 : S'_0 \rightarrow X$ sending the closed point s' of S' to x , there exists at least one (respectively, at most one, respectively, exactly one) S -morphism $u : S' \rightarrow X$ extending u_0 .

This statement shows in particular that, in Definition 1.1, one may restrict to local schemes S' , provided the functor under consideration is representable by a prescheme locally of finite presentation over S . Moreover, when S is locally noetherian, one may even restrict the preceding pointwise criterion to local artinian schemes S' , and the same restriction may therefore be made in Definition 1.1, provided that S is locally noetherian and that the functor in question is represented by a prescheme locally of finite type over S .

Remark 1.6. — Of course, given a morphism $f : X \rightarrow Y$ of preschemes, one says that f is smooth (respectively, unramified, respectively, étale) if f makes X a smooth (respectively, unramified, respectively, étale) Y -prescheme. When X and Y are S -preschemes and f is an S -morphism of finite presentation, these properties of f can be expressed immediately in terms of the morphism of functors

$$F, G : (\mathbf{Sch})^\circ / S \longrightarrow (\mathbf{Ens})$$

defined by f . Namely, f is smooth (respectively, unramified, respectively, étale) if, for every S -prescheme S' that is affine in the absolute sense and every subscheme $S'_0 \subset S'$ defined by a nilpotent ideal $\mathcal{J}$, the map

$$F(S') \longrightarrow F(S'_0) \times_{G(S'_0)} G(S')$$

deduced from the commutative square

$$\begin{array}{ccc} F(S') & \longrightarrow & G(S') \\ \downarrow & & \downarrow \\ F(S'_0) & \longrightarrow & G(S'_0) \end{array}$$

²The source prints $\mathcal{O}_{X,s}$; the local ring at the point of X is $\mathcal{O}_{X,x}$.

³Editor’s note: “residue extension” was replaced by “residue field.”

is surjective (respectively, injective, respectively, bijective). Equivalently, for every commutative square of morphisms of functors over S , where S' and S'_0 are as above and $i : S'_0 \rightarrow S'$ is the canonical immersion,

$$(Q) \quad \begin{array}{ccc} S'_0 & \xrightarrow{i} & S' \\ u_0 \downarrow & & \downarrow v \\ F & \xrightarrow{f} & G, \end{array}$$

there exists at least one (respectively, at most one, respectively, exactly one) morphism

$$\begin{array}{ccc} & & S' \\ & \swarrow u & \\ F & & \end{array}$$

that makes the two corresponding triangles commute:

$$u_0 = u \circ i \quad \text{and} \quad v = f \circ u.$$

This property for a homomorphism of functors $f : F \rightarrow G$ over S retains its meaning even when the functors under consideration are not representable. If it holds, the homomorphism $f : F \rightarrow G$ is said to be *formally smooth* (respectively, *formally unramified*, respectively, *formally étale*). Notice that this property depends only on the homomorphism of the underlying functors

$$(\mathbf{Sch})^\circ \longrightarrow (\mathbf{Ens})$$

defined by F and G (cf. Exposé I, 1.4.1), and not on the structure morphisms $F \rightarrow S$ and $G \rightarrow S$. An equivalent formulation of the preceding definition, somewhat easier to use in applications, is the following: a morphism of functors $F \rightarrow G$ over S is formally smooth (respectively, formally unramified, respectively, formally étale) if, for every S' over S and every morphism $S' \rightarrow G$, the functor

$$F' = F \times_G S'$$

over S' is formally smooth (respectively, formally unramified, respectively, formally étale).

Remark 1.7. — Under the hypotheses of 1.6, suppose that we are given a “point of F with values in a field k ,” that is, an element $x \in F(\mathrm{Spec} k)$, or equivalently a morphism

$$x : \mathrm{Spec} k \longrightarrow F.$$

We likewise say that f is *formally smooth* (respectively, *formally unramified*, respectively, *formally étale*) at x if the condition in the preceding definition is satisfied whenever S' is a local scheme and the morphism $S'_0 \rightarrow F$ in diagram (Q) above is “compatible with the marked points” in the following sense. Let k' denote the residue field of S' and S'_0 at their closed point. Then the diagram

$$\begin{array}{ccc} \mathrm{Spec}(k') & & \mathrm{Spec}(k) \\ j \downarrow & & \downarrow x \\ S'_0 & \xrightarrow{u_0} & F \end{array}$$

in which $j : \operatorname{Spec}(k') \rightarrow S'_0$ is the canonical immersion, must admit a completion to a commutative diagram

$$\begin{array}{ccc}
 & \operatorname{Spec}(k'') & \\
 \swarrow & & \searrow \\
 \operatorname{Spec}(k') & & \operatorname{Spec}(k) \\
 \downarrow & & \downarrow \\
 S'_0 & \xrightarrow{\quad} & F,
 \end{array}$$

where k'' is a field.⁴

When F and G are represented by S -preschemes X and Y , and the S -morphism $f : X \rightarrow Y$ is locally of finite presentation, this condition says precisely, by 1.5, that $f : X \rightarrow Y$ is smooth at the point of X that is the image of $\operatorname{Spec}(k)$ under $x : \operatorname{Spec}(k) \rightarrow X$.

Remark 1.8. — When the condition in the preceding definition is satisfied after restricting to local artinian schemes S' , we say that $F \rightarrow G$ is *infinitesimally smooth* (respectively, *infinitesimally unramified*, respectively, *infinitesimally étale*) at x . We say that $F \rightarrow G$ is infinitesimally smooth (respectively, infinitesimally unramified, respectively, infinitesimally étale) if it has the corresponding property at every point x ; equivalently, if the condition considered in 1.6 is satisfied whenever S' is local artinian.

This variant of the preceding notions is technically useful. Since it is weaker, it is often easier to verify, while it is nevertheless frequently sufficient to imply the stronger condition; this is the case, for example, when $F \rightarrow G$ is a morphism locally of finite presentation and G is represented by a locally noetherian prescheme.

Proposition 1.9. — *Let $f : X \rightarrow S$ be a smooth morphism of preschemes. Then $\Omega_{X/S}^1$ is a locally free module of finite type on X , and its rank at a point $x \in X$ is equal to the dimension of the fiber X_s , where $s = f(x)$, in a neighborhood of x .*

This dimension is called the *relative dimension of X over S at x* . Notice that it is zero (when f is smooth at x) if and only if f is étale at x . It can also be computed directly in terms of the functor F represented by X , as follows. Let ξ be a point of F with values in a field k , “localized at x ,” that is, a morphism

$$\operatorname{Spec}(k) \longrightarrow F = X$$

whose image is x . Consider the algebra

$$D(k) = k[t]/(t^2)$$

of dual numbers over k , regarded as an S -prescheme, and the map

$$F(D(k)) \longrightarrow F(k)$$

induced by the augmentation $D(k) \rightarrow k$. Finally, let $F(D(k), \xi)$ be the inverse image of $\xi \in F(k)$ under this map. This set is naturally endowed with the structure of a vector space over k . In fact, it is the vector-space dual of

$$\Omega_{X/S}^1 \otimes_{\mathcal{O}_{X,x}} k,$$

⁴The source says that k'' is “the spectrum of a field”; the displayed object $\operatorname{Spec}(k'')$ makes clear that k'' itself is the field.

and its dimension is the relative dimension of X over S at x .

To describe the vector-space law on $F(D(k), \xi)$ explicitly, it is useful to introduce, more generally and as in Exposé II, for every vector space V over k , the algebra

$$D_k(V) = k + V,$$

where V is an ideal of square zero, and to regard $F(D_k(V), \xi)$, the inverse image of ξ under

$$F(D_k(V)) \longrightarrow F(k),$$

as a covariant functor of V with values in **(Ens)**. It is then enough that this functor commute with products of two factors. This means that F carries certain amalgamated sums of a very special kind to fiber products; compare Exposé II. This condition always holds when F is representable. It follows that the sets $F(D_k(V), \xi)$, and in particular $F(D(k), \xi)$, are endowed with vector-space structures over k . Thus one can define the relative dimension of F over X at the “point” ξ under conditions appreciably more general than the representability of F .

In the present Exposé, the fact that certain functors to be described explicitly are representable by preschemes smooth over S will be used chiefly through the following result. It will be our technical means of passing from constructions over the completion of the local ring A of a point s of a noetherian scheme S to a local ring A' over A that is étale over A , and hence, in particular, of passing from s to neighboring points.

Proposition 1.10. — *Let $f : X \rightarrow S$ be a smooth morphism of preschemes, let $s \in S$, and let $x \in X$ lie over s , with $\kappa(x)$ a finite separable extension of $\kappa(s)$. Then there exists a subscheme $S' \subset X$, étale over S , passing through x .⁵ Thus one can find an étale morphism $S' \rightarrow S$, a point $s' \in S'$ above s whose residue-field extension is $\kappa(x)/\kappa(s)$, and an S -morphism $S' \rightarrow X$ carrying s' to x .*

To construct S' , simply choose sections $g_1, \dots, g_n$ of $\mathcal{O}_X$ on a neighborhood of x whose images at x form a regular system of parameters for the local ring of the fiber X_s at x . The subscheme S' defined by the g_i is then étale over S at x , and, after shrinking S' , it is étale over S .

We shall use 1.10 when $\kappa(x) = \kappa(s)$, that is, when x is rational over $\kappa(s)$, or equivalently when x may be regarded as a section of X_s over $\text{Spec } \kappa(s)$. In that case, 1.10 has the character of an extension theorem for sections, after an étale extension of the base. It takes a particularly simple form in the following special case.

Corollary 1.11. — *Under the hypotheses of 1.10, suppose that S is the spectrum of a henselian local ring and that $\kappa(x) = \kappa(s)$. Then there exists a section of X over S passing through x . It is uniquely determined if X is in fact étale over S at x .*

Indeed, since S is henselian, under the hypotheses of 1.10 the prescheme S' contains an open subscheme that is finite over S and whose fiber at s is reduced to x . Since this open subscheme is étale over S , it is isomorphic to S , which proves the assertion. Notice that when S is the spectrum of a complete local ring, 1.10 or 1.11 is more or less equivalent to the classical “Hensel’s lemma,” and one sometimes refers to the result by that name.

⁵The French re-edition prints “a subscheme S' of S , étale over S , passing through x .” Since $x \in X$, and both the equivalent formulation and the construction immediately below use $S' \rightarrow X$ and sections of $\mathcal{O}_X$, the intended reading is $S' \subset X$.

2. Examples of Formally Smooth Functors from the Theory of Groups of Multiplicative Type

We shall interpret, in the language introduced in the preceding number, the results stated in Exposé IX, no. 3, concerning infinitesimal extensions of a homomorphism from a group of multiplicative type. Those results are consequences of the vanishing of the Hochschild cohomology of such a group, established in Exposé I.

Proposition 2.1. — *Let S be a prescheme, let H be a group of multiplicative type over S , and let G be an S -group prescheme smooth over S . Consider the functor over S*

$$M_H = \mathrm{Hom}_{S\text{-gr}}(H, G).$$

Its value at an S' over S is the set of homomorphisms of S' -groups from $H_{S'}$ to $G_{S'}$. This functor is formally smooth over S .

Cf. Exposé IX, 3.6. More generally:

Corollary 2.2. — *Let S and G be as above, and consider a homomorphism $u : H_1 \rightarrow H_2$ of group schemes of multiplicative type over S . With the preceding notation this induces a morphism of functors over S*

$$M_{H_2} \longrightarrow M_{H_1}, \quad M_{H_i} = \mathrm{Hom}_{S\text{-gr}}(H_i, G),$$

given by $w \mapsto w \circ u$. This morphism is formally smooth.

Indeed, by the definitions in 1.6, this is equivalent to the following assertion. Suppose that S is affine and that $S_0 \subset S$ is a subscheme defined by a nilpotent ideal. Let

$$v : H_1 \longrightarrow G$$

be a homomorphism of S -groups, and let

$$w_0 : (H_2)_{S_0} \longrightarrow G_{S_0}$$

be a homomorphism of S_0 -groups such that

$$w_0 u_{S_0} = v_{S_0}.$$

Then there exists a homomorphism of S -groups

$$w : H_2 \longrightarrow G$$

extending w_0 and satisfying $wu = v$.

To see this, first extend w_0 to a homomorphism of S -groups

$$w' : H_2 \longrightarrow G,$$

which is possible by 2.1. Set

$$v' = w'u : H_1 \longrightarrow G.$$

By the hypothesis on w_0 , one has $v'_{S_0} = v_{S_0}$. Consequently, by Exposé VIII, 3.6, there exists an element $g \in G(S)$ whose image in $G(S_0)$ is the identity element and such that

$$v = \mathrm{int}(g)v'.$$

It follows that

$$v = (\mathrm{int}(g)w')u,$$

and it is therefore enough to take

$$w = \text{int}(g)w'.$$

Corollary 2.3. — *With the notation of 2.1, regard $M_H = M$ as a functor with operator group G , where G acts by*

$$(v, g) \mapsto \text{int}(g) \circ v.$$

Then the corresponding morphism

$$R : G \times_S M \longrightarrow M \times_S M,$$

defined by

$$(g, v) \mapsto (\text{int}(g) \circ v, v),$$

is formally smooth.

After a base change $S' \rightarrow S$, this is equivalent to the following assertion.

Corollary 2.4. — *With the notation of 2.1, let $v_1, v_2 : H \rightarrow G$ be two morphisms of S -groups, and let $\text{Transp}(v_1, v_2)$ be the subfunctor of G consisting of the g such that*

$$\text{int}(g) \circ v_1 = v_2.$$

Then this functor is formally smooth over S . In particular, if $v_1 = v_2 = v$, the functor $\text{Centr}(v)$, the subgroup of G consisting of the h such that $\text{int}(h)v = v$, is formally smooth over S .

N.B. The pair (v_1, v_2) may be regarded as a section over S of the second factor in the codomain of the morphism R in 2.3, and $\text{Transp}(v_1, v_2)$ as the inverse-image functor of that section under R .

The assertion in 2.4 is itself equivalent to the following. Suppose that S is affine and that $S_0 \subset S$ is a subscheme defined by a nilpotent ideal. If $g_0 \in G(S_0)$ satisfies

$$\text{int}(g_0)(v_1)_{S_0} = (v_2)_{S_0},$$

then g_0 extends to an element $g \in G(S)$ such that

$$\text{int}(g)v_1 = v_2.$$

To prove this, first extend g_0 to a section g' of G over S , which is possible because G is smooth over S , and put

$$v'_2 = \text{int}(g')v_1.$$

The morphisms v_2 and v'_2 have the same restriction over S_0 . Hence, by Exposé IX, 3.6, there exists $g'' \in G(S)$ inducing the identity section over S_0 and such that

$$v_2 = \text{int}(g'')v'_2.$$

Therefore

$$v_2 = \text{int}(g'') \text{int}(g')v_1 = \text{int}(g''g')v_1,$$

and it is enough to take $g = g''g'$.

Proposition 2.1 bis. — *Let S be a prescheme and let G be an S -group prescheme smooth over S . Consider the functor*

$$M : (\mathbf{Sch})^\circ / S \longrightarrow (\mathbf{Ens}),$$

defined by

$$M(S') = \left\{ \begin{array}{c} \text{multiplicative-type subgroups} \\ \text{of } G_{S'} \end{array} \right\}.$$

Then M is formally smooth over S .

Cf. Exposé IX, 3.6 bis.

Corollary 2.2 bis. — *Let n be an integer, and consider the morphism of functors*

$$\varphi_n : M \longrightarrow M$$

defined by

$$\varphi_n(H) = {}_nH = \ker(n \cdot \text{id}_H).$$

Then φ_n is formally smooth. For every integer $p > 0$, let M_p denote the subfunctor of M for which $M_p(S')$ is the set of multiplicative-type subgroups H of $G_{S'}$ satisfying $pH = H$. Then the morphism induced by φ_n

$$M_{np} \longrightarrow M_n$$

is formally smooth.

The second assertion is trivially contained in the first and is included only for the convenience of a later reference. The proof of the first assertion is analogous to that of 2.2, now invoking Exposé IX, 3.6 bis.

Corollary 2.3 bis. — *With the notation of 2.1 bis, regard M as a functor with operator group G , where G acts by*

$$(g, H) \longmapsto \text{int}(g)(H).$$

Then the corresponding morphism

$$R : G \times_S M \longrightarrow M \times_S M,$$

defined by

$$(g, v) \longmapsto (\text{int}(g)v, v),$$

is formally smooth.

This is equivalent to the following assertion.

Corollary 2.4 bis. — *With the notation of 2.1 bis, let H_1 and H_2 be two multiplicative-type subgroups of G , and let $\text{Transp}_G(H_1, H_2)$ be the subfunctor of G consisting of the g such that*

$$\text{int}(g)H_1 = H_2.$$

Then this functor is formally smooth over S . In particular, if $H_1 = H_2 = H$, the subfunctor $\text{Norm}_G(H)$ of G , the normalizer of H , is formally smooth over S .

The proof is analogous to that of 2.4, again invoking Exposé IX, 3.6 bis.

Proposition 2.5. — *Let S be a prescheme, let G be an S -group prescheme smooth over S , let K be an S -group subprescheme that is smooth over S or of multiplicative type, let H be an S -group prescheme of multiplicative type, and let $u : H \rightarrow G$ be a homomorphism of S -groups. Denote by $\text{Transp}_G(u, K)$ the subfunctor of G whose value at an S' over S consists of the elements $g \in G(S')$ for which*

$$\text{int}(g)u_{S'} : H_{S'} \longrightarrow G_{S'}$$

factors through $K_{S'}$. Then this functor is formally smooth over S .

The proof is analogous to that of 2.2, invoking Exposé IX, 3.6 and Exposé X, 2.1, the latter in the case where K is of multiplicative type.

3. Auxiliary Representability Results

Proposition 3.1. — *Let $F \rightarrow S$ be a functor over the prescheme S . The following conditions are equivalent:*

- (i) *F is representable and $F \rightarrow S$ is an open immersion (one also says simply that $F \rightarrow S$ is an open immersion).*
- (ii) *F is a sheaf for the faithfully flat quasi-compact topology, “commutes with filtered direct limits of rings,” and $F \rightarrow S$ is a monomorphism. Moreover, the following condition holds: for every local prescheme S' over S , with residue field k , and every S -morphism*

$$\mathrm{Spec}(k) = S'_0 \longrightarrow F,$$

*there exists an S -morphism $S' \rightarrow F$ extending it.*⁶

- (iii) *Suppose that S is locally noetherian. Then F is a sheaf for the faithfully flat quasi-compact topology, commutes with filtered direct limits of rings, commutes with adic inverse limits of local artinian rings, and $F \rightarrow S$ is a monomorphism and is infinitesimally étale (cf. 1.8).*

Let us first make two points of terminology precise.

Remark 3.2. — A functor⁷ F over S is said to *commute with direct limits of rings* (understood to mean filtered direct limits) if the following holds. Let $(S'_i)_{i \in I}$ be any filtered inverse system of affine S -schemes lying over an affine open subscheme of S , and let A'_i be their rings. Then the natural homomorphism

$$(*) \quad \varinjlim_i F(S'_i) \longrightarrow F(S'), \quad S' = \mathrm{Spec}(A'), \quad A' = \varinjlim_i A'_i$$

is bijective.

Notice that S' is precisely the inverse limit of the S'_i in the category of preschemes, and even in the category of all ringed spaces. Thus the condition under consideration has the character of a right-exactness condition—commutation with certain direct limits in $(\mathbf{Sch})^\circ/S$ —just as does the condition of being a sheaf for a topology.

Care must be taken that this condition is essentially relative: it involves the morphism $F \rightarrow S$, not merely the functor

$$F : (\mathbf{Sch})^\circ \longrightarrow (\mathbf{Ens}).$$

More precisely, in $(*)$, $F(S'_i)$ and $F(S')$ mean $\mathrm{Hom}_S(S'_i, F)$ and $\mathrm{Hom}_S(S', F)$. Consequently, when F is representable, the condition says that F is locally of finite presentation over S . We used several times in the preceding two exposés the fact that a functor represented by an S -prescheme locally of finite presentation commutes with direct limits of rings.

Remark 3.3. — A functor F over S is said to *commute with adic inverse limits of local artinian rings* if the following condition holds. For every S' over S that is the spectrum of a complete noetherian local ring A' , set

$$S'_n = \mathrm{Spec}(A' / \mathrm{rad}(A')^{n+1}).$$

⁶The French re-edition prints $S' \rightarrow S$. Since this morphism must extend the displayed morphism $S'_0 \rightarrow F$, its intended target is F .

⁷Editor’s note: here and in Remark 3.3, the functor is contravariant.

Then the natural map

$$(**) \quad F(S') \longrightarrow \varprojlim_n F(S'_n)$$

is required to be bijective.

This condition has the character of a left-exactness condition and is satisfied whenever F is representable. In contrast to commutation with direct limits of rings, it is intrinsic to F as an object of $\widehat{\mathbf{Sch}}$; that is, it does not involve the morphism $F \rightarrow S$.

Remark 3.4. — Let F be a functor over S that is a sheaf for the Zariski topology, or, as one also says, is “of local nature.” It is enough for this that F be a sheaf for a finer topology, such as the faithfully flat quasi-compact topology. Let (S_i) be an open covering of S . A straightforward gluing argument shows that F is representable if and only if each

$$F_i = F \times_S S_i$$

is representable; this allows one, for example, to reduce to the case where S is affine.

Suppose that the functor F , of local nature, commutes with direct limits of rings. Then F is representable if and only if its restriction to the category of preschemes locally of finite presentation over S is representable. Necessity was noted in 3.2. Sufficiency amounts to the following assertion: if X is a prescheme locally of finite presentation over S , and $X \rightarrow F$ is a morphism such that, for every S' locally of finite presentation over S , the induced map

$$\mathrm{Hom}_S(S', X) \longrightarrow \mathrm{Hom}_S(S', F)$$

is bijective, then $X \rightarrow F$ is an isomorphism. This follows readily from the fact that X and F are both functors of local nature that commute with direct limits of rings.

Proof of Proposition 3.1. The implications (i) $\Rightarrow$ (ii) and (i) $\Rightarrow$ (iii) are immediate. We prove the converse implications.

Assume (ii). Let U be the set of points $s \in S$ for which the canonical monomorphism

$$\mathrm{Spec} \kappa(s) \longrightarrow S$$

factors through F . By the last condition in (ii), for every $s \in U$, the canonical map $\mathrm{Spec} \mathcal{O}_{S,s} \rightarrow S$, equivalently its factorization through U , factors through F . The local ring $\mathcal{O}_{S,s}$ is the direct limit of the rings of affine neighborhoods of s . Since F commutes with direct limits of rings, for every $s \in U$ ⁸ there is an open neighborhood U^s such that the canonical immersion $U^s \rightarrow S$ factors through F . It follows that $U^s \subset U$, and hence that U is open.

Because $F \rightarrow S$ is a monomorphism and F is of local nature, the S -morphisms $U^s \rightarrow F$ agree on the intersections $U^s \cap U^{s'}$, for $s, s' \in U$, and therefore glue to an S -morphism $U \rightarrow F$. It remains to prove that this morphism is an isomorphism, or equivalently that every S -morphism $S' \rightarrow F$ factors uniquely through U , where S' is an S -prescheme. Since both $F \rightarrow S$ and $U \rightarrow S$ are monomorphisms, it is equivalent to say that the structure morphism $S' \rightarrow S$ factors through U .

⁸The French re-edition prints $s \in S$ at this occurrence; the argument and the conclusion $U^s \subset U$ require $s \in U$.

By locality, we are reduced to the case in which S' is the spectrum of a field and hence has a single point s' . Let s be the point of S below s' . We claim that the given S -morphism $S' \rightarrow F$ factors through

$$\mathrm{Spec} \kappa(s) = S_0 \longrightarrow F.$$

This implies $s \in U$ and proves the assertion. Since $S' \rightarrow S_0$ is an fpqc covering and F is a sheaf for the fpqc topology, it is enough that the two composites

$$S'' = S' \times_{S_0} S' \rightrightarrows S' \longrightarrow F$$

coincide. They do coincide because $F \rightarrow S$ is a monomorphism. This proves (ii) $\Rightarrow$ (i).

Now assume (iii), with S locally noetherian. It is enough to prove the last condition of (ii); by the preceding proof, it is enough to do so when

$$S' = \mathrm{Spec} \mathcal{O}_{S,s}$$

for some $s \in S$. Put

$$A = \mathcal{O}_{S,s}, \quad A_n = A/\mathfrak{m}^{n+1}, \quad S_n = \mathrm{Spec}(A_n).$$

Because $F \rightarrow S$ is infinitesimally smooth, the given morphism $S'_0 \rightarrow F$ extends to morphisms $S_n \rightarrow F$. Since $F \rightarrow S$ is a monomorphism, these morphisms form an element of $\varprojlim_n F(S_n)$. Since F commutes with adic inverse limits of local artinian rings, they arise from a morphism

$$\mathrm{Spec}(\widehat{A}) = \widehat{S}' \longrightarrow F.$$

Finally, $F \rightarrow S$ is a monomorphism, F is an fpqc sheaf, and $\widehat{S}' \rightarrow S'$ is an fpqc covering. The last morphism therefore descends to $S' \rightarrow F$, completing the proof of (iii) $\Rightarrow$ (ii). $\square$

Proposition 3.5. — *Let S be a locally noetherian prescheme, let $F \rightarrow S$ be a functor over S , and let $(X_i, u_i)_{i \in I}$ be a family of S -morphisms $u_i : X_i \rightarrow F$, where the X_i are preschemes locally of finite type over S . Suppose that the following conditions hold:*

- (a) *F is a sheaf for the faithfully flat quasi-compact topology, commutes with direct limits of rings, and commutes with adic inverse limits of local artinian rings.*
- (b) *The morphisms $u_i : X_i \rightarrow F$ are monomorphisms and are infinitesimally étale (cf. 1.8).*
- (c) *The family (u_i) is “set-theoretically surjective.”*

Under these conditions, F is representable by a prescheme locally of finite type over S . Moreover, the u_i are open immersions and the family (X_i) is an open covering of F .

Remark 3.6. — Proceeding as at the end of Remark 1.7, one defines, for every functor

$$F : (\mathbf{Sch})^\circ \longrightarrow (\mathbf{Ens}),$$

an “underlying set” $\mathrm{ens}(F)$: it is the quotient of the set of field-valued points of F by the equivalence relation described in 1.7. When F is represented by X , this recovers the underlying set of X .

Plainly, $\mathrm{ens}(F)$ depends functorially on F . Thus, if $G \rightarrow F$ is a morphism of functors, we say that it is *set-theoretically surjective* if the induced map

$$\mathrm{ens}(G) \longrightarrow \mathrm{ens}(F)$$

is surjective. Equivalently, every point of F with values in a field k “comes from” a point of G with values in a suitable extension of k . This definition extends immediately to a family of morphisms $G_i \rightarrow F$, and thereby specifies the meaning of condition (c).

Proof of Proposition 3.5. For $(i, j) \in I \times I$, put

$$X_{i,j} = X_i \times_F X_j$$

and consider the projections

$$v_{i,j} : X_{i,j} \longrightarrow X_i, \quad w_{i,j} : X_{i,j} \longrightarrow X_j.$$

We claim that these morphisms are representable by open immersions. Apply criterion 3.1(iii). The functor $X_{i,j}$ satisfies the three exactness conditions: it is a sheaf, it commutes with direct limits of rings, and it commutes with adic inverse limits of local artinian rings. Indeed, F , X_i , and X_j satisfy these conditions, and the conditions are stable under finite inverse limits, in particular under fiber products. Since every $X_i \rightarrow F$ is a monomorphism, both projections are monomorphisms, each being obtained from one of these maps by base change. Finally, being infinitesimally étale is also preserved by base change. Thus the hypotheses of 3.1(iii) are satisfied.

We may now use the X_i , $X_{i,j}$, $v_{i,j}$, and $w_{i,j}$ to construct in the usual way an S -prescheme X , in which the X_i are identified with open subschemes, the $X_{i,j}$ with the intersections $X_i \cap X_j$, and $v_{i,j}$, $w_{i,j}$ with the canonical immersions.

Put⁹

$$Y = \coprod_i X_i.$$

Then X is also the quotient of Y by the equivalence relation

$$R = \coprod_{i,j} X_{i,j} \rightrightarrows Y,$$

whose two projections v, w are defined by the $v_{i,j}$ and $w_{i,j}$, respectively.

More precisely, because F is an fpqc sheaf, the morphisms $u_i : X_i \rightarrow F$ induce a morphism $u : Y \rightarrow F$, and

$$R = Y \times_F Y$$

is exactly the equivalence relation defined by u . The quotient $X = Y/R$ is also a quotient in the category of fpqc sheaves, and even in the category of Zariski sheaves; this follows directly from the definitions of quotient and sheaf. Consequently, u factors uniquely through a morphism

$$\bar{u} : X \longrightarrow F,$$

and this morphism is a monomorphism.

It remains to show that $\bar{u}$ is an isomorphism. Since F is of local nature, we may assume that S is affine. Since F also commutes with direct limits of rings, it is enough, by 3.4, to verify that, for every affine T of finite type over S , every morphism $T \rightarrow F$ factors through X . Consider

⁹The French re-edition labels this coproduct X and later writes $R = X \times_F X$, while simultaneously using $R \rightrightarrows Y$ and $X = Y/R$. The coherent notation, adopted here, is $Y = \coprod_i X_i$ and $R = Y \times_F Y$.

$$G = X \times_F T \longrightarrow T.$$

It is enough to prove that this morphism is an isomorphism. Since T is noetherian, the argument above shows that it is an open immersion. Here one uses that $X \rightarrow F$ is infinitesimally étale, which follows immediately because the induced morphisms $u_i : X_i \rightarrow F$ are infinitesimally étale. By hypothesis, $X \rightarrow F$ is set-theoretically surjective, and this condition is stable under base change. Hence $G \rightarrow T$ is set-theoretically surjective. Being also an open immersion, it is an isomorphism. $\square$

Proposition 3.7. — *Let S be a locally noetherian prescheme, let I be a directed partially ordered index set, let $(T_i)_{i \in I}$ be a projective system of S -preschemes locally of finite type, let*

$$T = \varprojlim_{i \in I} T_i$$

be the projective-limit functor, let F be a functor over S , and let $u : F \rightarrow T$ be an S -morphism. Suppose that the following conditions hold:

- (a) *F is a sheaf for the faithfully flat quasi-compact topology, commutes with inductive limits of rings, and commutes with adic projective limits of local artinian rings.*
- (b) *The morphism $u : F \rightarrow T$ is a monomorphism.*
- b' *The morphism $u : F \rightarrow T$ is infinitesimally étale.*
- (c) *For every point ξ of F with values in the spectrum of a field k , let $\xi_i \in T_i(\text{Spec } k)$ be its image and let t_i be the corresponding point of T_i . There exists $i \in I$ such that, for every $j \geq i$, the transition morphism*

$$p_{i,j} : T_j \longrightarrow T_i$$

is étale at t_j .

- (d) *For every prescheme X locally of finite type over S and every S -morphism $X \rightarrow F$, the set of points $x \in X$ at which this morphism is infinitesimally étale is open.*

Under these conditions, F is representable by a prescheme locally of finite type over S .

Let us note at once that, in the case to be considered in the next section, conditions (c) and (d) will be verified by means of the following corollary.

Corollary 3.8. — *Assuming (a), (b), and (b'), conditions (c) and (d) are implied by the following conditions:*

- c' *The T_i are smooth over S , and the transition morphisms $p_{i,j} : T_j \rightarrow T_i$ are smooth.*
- d' *For every point ξ of F with values in the spectrum of a field k , let $t_i(\xi)$ be the point of T_i defined by ξ , let $d_i(\xi)$ be the relative dimension of T_i over S at $t_i(\xi)$, and put*

$$d(\xi) = \sup_i d_i(\xi).$$

Then:

- 1° *for every ξ as above, $d(\xi) < +\infty$; and*

2° for every prescheme X locally of finite type over S and every S -monomorphism $v : X \rightarrow F$, the function

$$x \longmapsto d(\xi_x)$$

on X is locally constant, where, for $x \in X$, ξ_x denotes the point of F with values in $\kappa(x)$ induced by v .

Proof of Proposition 3.7. Work under condition (c). Let t be the element of $\text{ens}(F)$ defined by ξ (cf. 3.6), and put

$$\mathcal{O}_t = \varinjlim_i \mathcal{O}_{T_i, t_i}.$$

Using condition (c), one sees readily that $\mathcal{O}_t$ is a noetherian local ring (EGA 0_{IV}, 10.3.1.3). Its residue field $\kappa(t)$ is the inductive limit of the residue fields $\kappa(t_i)$, and k is an extension of it. If η denotes $\text{Spec } \kappa(t)$, and ξ denotes $\text{Spec } k$, there is a commutative diagram

$$\begin{array}{ccc} \xi & \longrightarrow & F \\ \downarrow & & \downarrow \\ \eta & \longrightarrow & T, \end{array}$$

whose definition is evident and is left to the reader. Since $\xi \rightarrow \eta$ is a covering for the fpqc topology, F is a sheaf for that topology by (a), and $F \rightarrow T$ is a monomorphism by (b), the morphism $\eta \rightarrow T$ factors uniquely through a morphism $\eta \rightarrow F$. We now record the following lemma.

Lemma 3.9. — *Under the hypotheses of 3.7, let*

$$T' = \text{Spec}(A) \quad \text{and} \quad T'_0 = \text{Spec}(k(A))$$

with T' a noetherian local scheme, and let $i : T'_0 \rightarrow T'$ be the canonical immersion. Suppose that a commutative square of morphisms is given:

$$\begin{array}{ccc} T'_0 & \longrightarrow & F \\ i \downarrow & & \downarrow \\ T' & \longrightarrow & T. \end{array}$$

Then there exists a unique T -morphism $T' \rightarrow F$.

The proof is the proof of 3.1, (iii) $\Rightarrow$ (ii), in which the representability of the S occurring in that statement was not used. One uses the facts that F is a sheaf for the fpqc topology, that it commutes with adic projective limits of local artinian rings, here the rings

$$A/\text{rad}(A)^{n+1},$$

and that $F \rightarrow T$ is both a monomorphism and infinitesimally étale.

We apply Lemma 3.9 with

$$T' = \text{Spec}(\mathcal{O}_t), \quad T'_0 = \text{Spec}(\kappa(t)).$$

We have just constructed $T'_0 \rightarrow F$, and the canonical homomorphisms

$$\mathcal{O}_{T_i, t_i} \longrightarrow \mathcal{O}_t$$

define a projective system of morphisms $\mathrm{Spec}(\mathcal{O}_t) \rightarrow T_i$, hence a canonical morphism $\mathrm{Spec}(\mathcal{O}_t) \rightarrow T$. The corresponding square (x) plainly commutes: by the definition of T , it is enough to check this after replacing T by T_i . We therefore obtain a unique T -morphism

$$v' : T' = \mathrm{Spec}(\mathcal{O}_t) \longrightarrow F.$$

Since F commutes with inductive limits of rings, this morphism factors, for sufficiently large i , through

$$v'_i : T'_i = \mathrm{Spec}(\mathcal{O}_{T_i, t_i}) \longrightarrow F.$$

For such an i , one has

$$T' \xrightarrow{\sim} T'_i, \quad \text{that is,} \quad \mathcal{O}_{T_i, t_i} \xrightarrow{\sim} \mathcal{O}_t.$$

Indeed, $T' \rightarrow T'_i$ is faithfully flat and quasi-compact, hence an effective epimorphism, so it is enough to show that it is a monomorphism. Suppose that two morphisms $R \rightrightarrows T'$, with R representable, have the same composite with $T' \rightarrow T'_i$. Their composites with $T' \rightarrow T$, which factors through $T' \rightarrow T'_i$, are then equal. Hence their composites with $T' \rightarrow T_j$, and therefore with $T' \rightarrow T'_j$, are equal for every j . The two original morphisms are equal because T' is the projective limit of the T'_j in the category **(Sch)**.

Consider the diagram

$$\begin{array}{ccc} T' & \xrightarrow{v'} & F \\ \downarrow & \nearrow v'_i & \downarrow \\ T'_i & \xrightarrow{\quad} & T_i, \end{array}$$

where the unlabeled arrows are the evident ones. The square and the upper triangle commute. Since $T' \rightarrow T'_i$ is an epimorphism, the lower triangle also commutes.

The ring $\mathcal{O}_{T_i, t_i}$ is the inductive limit of the rings of affine open neighborhoods of t_i in T_i . Since F commutes with inductive limits of rings, v'_i comes from a morphism

$$v_t : U_t \longrightarrow F$$

defined on an open neighborhood U_t of t_i in T_i . After shrinking this neighborhood if necessary, v_t is a T_i -morphism. Indeed, T_i , being locally of finite type over S , also commutes with inductive limits of rings. The composite of v_t with $F \rightarrow T_i$ is a monomorphism, and therefore v_t is a monomorphism.

We claim that v_t is infinitesimally étale at t_i . Since $F \rightarrow T$ is infinitesimally étale, this is equivalent to the composite

$$T'_i \xrightarrow{v'_i} F \longrightarrow T$$

being infinitesimally étale at t_i , or to the induced morphism $T'_i \rightarrow T$ being infinitesimally étale. Since $T' \xrightarrow{\sim} T'_i$, it is in turn equivalent to $T' \rightarrow T$ being infinitesimally étale. This is immediate, because that morphism is the projective limit of the infinitesimally étale morphisms $T'_j \rightarrow T_j$.

Apply condition (d), which has not yet been used. It follows that v_t is infinitesimally étale in a neighborhood of t_i . After replacing U_t by a smaller open, we may therefore suppose that v_t is a monomorphism that is infinitesimally étale.

As t varies in $\mathrm{ens}(F)$, the family of morphisms v_t satisfies Proposition 3.5, which gives the conclusion of Proposition 3.7.

Proof of Corollary 3.8. Assume conditions (c') and (d'). With the notation of (d'), $d_i(\xi)$ is constant for all sufficiently large i . Consequently, the relative dimension of T_j over T_i at t_j , which is

$$d_j(\xi) - d_i(\xi),$$

is zero. Thus $T_j \rightarrow T_i$ is étale at t_j , which proves condition (c).

The preceding proof now gives, for each $t \in \text{ens}(F)$, an index i , an open neighborhood U_t of t_i , and a morphism $U_t \rightarrow F$ that is a monomorphism and is infinitesimally étale at t_i . It remains to show that this morphism is infinitesimally étale throughout a neighborhood of t_i . In condition (d'), 2° , take $X = U_t$. Replacing U_t , if necessary, by the connected component of t_i in U_t , we may suppose that

$$d(\xi_x) = d(\xi_{t_i}) \quad \text{for every } x \in U_t.$$

Now $d(\xi_{t_i})$ is the relative dimension of T_i over S at t_i . Since that relative dimension remains constant on U_t , we also have, for every $x \in U_t$,

$$(*) \quad d(\xi_x) = \text{the relative dimension of } U_t \text{ over } S \text{ at } x.$$

On the other hand, using the notation of the proof of 3.7, consider any local noetherian prescheme

$$R' = \text{Spec}(A), \quad R'_0 = \text{Spec}(k(A)).$$

Every morphism $R' \rightarrow F$ whose induced point $R'_0 \rightarrow F$ represents $t \in \text{ens}(F)$, that is, sends the closed point of R' to t , factors uniquely through T' . The proof is the proof of 3.1 or of Lemma 3.9: one uses that $T' \rightarrow F$ is a monomorphism infinitesimally étale at t , and that T' is an fpqc sheaf commuting with adic projective limits.

Apply this result to the morphism

$$\text{Spec}(\mathcal{O}_{U_t, x}) = R' \longrightarrow F$$

induced by v_t , and to the point $t_x \in \text{ens}(F)$ that is the image of the closed point of R' , or equivalently the image of x under v_t . We obtain a factorization

$$\text{Spec}(\mathcal{O}_{U_t, x}) \longrightarrow \text{Spec}(\mathcal{O}_{t_x}) \longrightarrow F.$$

The second arrow is infinitesimally étale at t_x . To prove that the composite is infinitesimally étale at x , it is therefore enough to prove the same for the first arrow at x .

By (x) above and formula (*), the first arrow is induced by a local S -homomorphism between the localizations, at x and t_x , of the smooth S -preschemes U_t and U_{t_x} , which have the same relative dimension d over S at those points. By EGA I, 6.5.1, this local homomorphism is induced by an S -morphism

$$w : U \longrightarrow V,$$

where U is an open neighborhood of x in U_t and $V = U_{t_x}$. It remains to prove that w is étale at x . Moreover, U and V carry monomorphisms to F , and, after shrinking U once more, w is an F -morphism. Hence w is a monomorphism. It is now enough to prove the following lemma.

Lemma 3.10. — *Let U and V be two smooth S -preschemes of the same relative dimension d over S , and let $w : U \rightarrow V$ be an S -morphism that is a monomorphism. Then w is an open immersion, and hence is étale.*

By SGA 1, I, 5.7, we reduce to the case in which S is the spectrum of a field, which may be assumed algebraically closed. By SGA 1, I, 5.1, it is enough to prove that w is étale, and it is enough to do so at the closed points of U . Let x be such a point and put $y = w(x)$. Choose a regular system of parameters $f_1, \dots, f_d$ of $\mathcal{O}_{V,y}$. Then

$$A = \mathcal{O}_{U,x} / \sum_i f_i \mathcal{O}_{U,x}$$

is the trivial extension of

$$k = \mathcal{O}_{V,y} / \sum_i f_i \mathcal{O}_{V,y};$$

that is, $A \simeq k$. Indeed, since w is a monomorphism, so is the structural morphism

$$\mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(k)$$

obtained from it by base change. Since $\mathcal{O}_{U,x}$ is a regular local ring of dimension d , the f_i form a regular system of parameters of that ring. It follows immediately that w is étale at x . This proves Lemma 3.10, and hence Corollary 3.8.

Corollary 3.11. — *Under the hypotheses of 3.8, for every quasi-compact open subfunctor U of F that is separated over S , there exists $i \in I$ such that, for every $j \geq i$, the morphism*

$$u_j|_U : U \longrightarrow T_j$$

is an open immersion. In particular, if the T_i are quasi-affine over S , then every open subfunctor U of F that is quasi-compact over S , that is, of finite type over S , is quasi-affine over S .

The proof of Proposition 3.7 shows that for every $t \in \mathrm{ens}(F)$ there exists $i \in I$ such that $u_i : F \rightarrow T_i$ is a local isomorphism at t , and then u_j is a local isomorphism at that point for every $j \geq i$. By quasi-compactness, i may be chosen independently of $x \in U$. It remains to prove that, for sufficiently large i ,

$$u_i|_U : U \longrightarrow T_i$$

is a monomorphism.

Since $U \rightarrow T$ is a monomorphism, the intersection of the equivalence relations

$$U \times_{T_i} U \subset U \times_S U$$

is the diagonal. The prescheme $U \times_S U$ is noetherian, and the $U \times_{T_i} U$ are closed subschemes. It follows that one of them is already the diagonal; equivalently, $u_i|_U$ is a monomorphism. This proves the first assertion of Corollary 3.11, and the second follows immediately.

Proposition 3.12. — *Let S be a prescheme and G an affine S -group prescheme.*

(a) *Let*

$$F : (\mathbf{Sch})^\circ / S \longrightarrow (\mathbf{Ens})$$

be the functor such that, for every T over S ,

$$F(T) = \left\{ \begin{array}{c} \text{multiplicative-type subgroups of } G_T \\ \text{that are finite over } T \end{array} \right\}.$$

Suppose that S is locally noetherian or that G is of finite presentation over S . Then F is representable and affine over S . If G is of finite presentation over S , then F is locally of finite presentation over S .

- (b) Let H be a finite S -group prescheme of multiplicative type. Then $\mathrm{Hom}_{S\text{-gr}}(H, G)$ is representable and affine over S . If G is of finite type (respectively, of finite presentation) over S , then the same is true of $\mathrm{Hom}_{S\text{-gr}}(H, G)$.

Remark 3.13. — Apart from the assertion that $\mathrm{Hom}_{S\text{-gr}}$ is affine, and from the case in which G is of finite presentation over S , which is the case that will suffice for us, 3.12 is an immediate consequence of the theory of Hilbert schemes (A. Grothendieck, *Techniques de construction et théorèmes d'existence en Géométrie Algébrique: IV. Les Schémas de Hilbert*, Séminaire Bourbaki, May 1961, no. 221). It is enough, in fact, that G be quasi-projective over S . In case (a) one can also represent the larger functor

$$F_0(T) = \left\{ \begin{array}{l} \text{group subpreschemes of } G_T \text{ that are flat, proper,} \\ \text{and of finite presentation over } T \end{array} \right\}.$$

The canonical monomorphism $F \rightarrow F_0$ is an “open immersion,” as follows from the criterion in 3.1 and Exposé X, 4.7 (b). Thus the representability of F_0 implies that F is representable by an open subprescheme of F_0 . In case (b) it is enough to assume that H is projective and of finite presentation over S . In both cases one obtains a functor locally of finite presentation over S .

In the present Exposé, 3.12 is only a technical lemma used to prove a key result in the next section. We shall therefore sketch an easy direct proof that does not use Hilbert schemes.

We first prove (b). It is enough to prove that $\mathrm{Hom}_S(H, G)$ is representable, independently of any group structure on H and G , and that it has the additional properties asserted for $\mathrm{Hom}_{S\text{-gr}}(H, G)$. Indeed, applying the same result to $\mathrm{Hom}_S(H \times_S H, G)$, the subfunctor $\mathrm{Hom}_{S\text{-gr}}(H, G)$ of $\mathrm{Hom}_S(H, G)$ can be expressed by a finite inverse limit—in fact, by fiber products—involving G , $\mathrm{Hom}_S(H, G)$, and $\mathrm{Hom}_S(H \times_S H, G)$. We leave this expression to the reader.

Write

$$H = \mathrm{Spec}(B),$$

where B is a sheaf of algebras on S that is locally free as a sheaf of modules; this is the only hypothesis on H that the argument uses. Since the representability question is local on S , we may suppose that S is affine, with ring A . We may also write $G = \mathrm{Spec}(C)$, where C is an A -algebra.

When $G = S[t] = G_0$, with t an indeterminate, the Hom functor is simply

$$T \longmapsto \Gamma(T, B_T).$$

Since B is locally free, this functor is represented by the vector bundle $V(B^\vee)$, where $B^\vee$ is the dual module sheaf. When $G = S[(t_i)_{i \in I}] = G_0^I$, for a not necessarily finite family of indeterminates, it follows that¹⁰

$$\mathrm{Hom}_S(H, G) = \mathrm{Hom}_S(H, G_0)^I = V(B^\vee)^I = V(B^{\vee(I)}).$$

In general G is isomorphic to a closed subscheme of a scheme of the form $S[(t_i)_{i \in I}]$; equivalently, C is a quotient of an A -algebra $A[(t_i)_{i \in I}]$. Let (F_j) be a family of generators

¹⁰The source display prints $\mathrm{Hom}_S(H, G)^I = V(B^\vee)^I = V(B^{\vee(I)})$. Since the preceding sentence sets $G = G_0^I$, the displayed equality requires the explicit factor $\mathrm{Hom}_S(H, G) = \mathrm{Hom}_S(H, G_0)^I$.

of the ideal by which one takes the quotient. After covering S by smaller affine open subsets, we may assume that B is free of rank n . Choose a basis $(e_k)_{1 \leq k \leq n}$ of B . Writing

$$x_i = \sum_{k=1}^n x_{ik} e_k$$

with indeterminate coefficients x_{ik} in an unspecified A -algebra A' , expand $F_j((x_i))$ in this basis. For every F_j this gives n polynomials

$$F_{j,k} \quad (1 \leq k \leq n)$$

in the variables $(x_{i,k})_{i \in I, 1 \leq k \leq n}$, with coefficients in A . It follows at once that $\mathrm{Hom}_S(H, G)$ is represented by the spectrum of the quotient of the polynomial ring $A[(x_{i,k})]$ by the ideal generated by the $F_{j,k}$. This proves (b).

We now prove (a). For every ordinary finite commutative group M , let F_M be the subfunctor of F obtained by restricting to those subgroups of G_T that are of multiplicative type and of type M . A gluing argument like the one used in 3.5 (which we should have stated after decomposing the problem a little further) shows that it is enough to prove that each F_M is representable. Then F is represented by the disjoint sum of the F_M , with M ranging over the isomorphism classes of finite commutative groups. Indeed, F is the corresponding sum in the category of sheaves.

Fix M and write F in place of F_M . Put $H = D_S(M)$, and let

$$F' = \mathrm{Imm}_{S\text{-gr}}(H, G) \subset \mathrm{Hom}_{S\text{-gr}}(H, G)$$

be the subfunctor whose value at T is the set of homomorphisms of T -groups $H_T \rightarrow G_T$ that are closed immersions. By (b), $\mathrm{Hom}_{S\text{-gr}}(H, G)$ is represented by an affine S -prescheme. Using Exposé IX, 6.8, one sees immediately that F' is represented by an open-and-closed subprescheme of it; in particular, F' is affine over S .

Consider the canonical morphism $F' \rightarrow F$, which assigns to every monomorphism $H_T \rightarrow G_T$ its image group. Directly from the definitions, F' is thereby a principal homogeneous bundle over F , in the category of fpqc sheaves, under

$$\Gamma_F = \Gamma_S \times_S F, \quad \Gamma = \mathrm{Aut}_{\mathrm{gr}}(M).$$

It follows by descent that $F' \rightarrow F$ is representable: for every morphism $T \rightarrow F$ with T representable, the prescheme $T \times_F F'$ is representable over T , in fact by a principal Galois bundle under Γ .

By Exposé IV, 4.6.6, F is therefore representable if and only if the quotient F'/F'' , where $F'' = F' \times_F F'$, exists in **(Sch)** and is universally effective for faithfully flat quasi-compact morphisms. Equivalently, the quotient F'/Γ must exist and be universally effective for such morphisms. Since F' is affine over S , Exposé V, 4.1 shows that this condition is satisfied. This proves the representability of F in (a).

Finally, suppose that G is of finite presentation over S . The additional assertion follows immediately from the preceding proof: by (b), F' is then locally of finite presentation over S , so that Exposé V may be applied.¹¹

¹¹Editor's note: the source marks this final reference to Exposé V as needing verification or further specification.

4. The Scheme of Multiplicative-Type Subgroups of a Smooth Affine Group

The principal result of the present Exposé is the following.

Theorem 4.1. — *Let S be a prescheme, let G be an S -group prescheme smooth and affine over S , and let*

$$F: (\mathbf{Sch})^\circ/S \longrightarrow (\mathbf{Ens})$$

be the functor defined by

$$F(T) = \{\text{multiplicative-type subgroups of } G_T\}.$$

Then F is representable, smooth, and separated over S .

Let us record at once the following variant.

Corollary 4.2. — *Let G and H be two S -group preschemes, with G smooth and affine over S , and H of multiplicative type and of finite type. Then $\text{Hom}_{S\text{-gr}}(H, G)$ is representable, smooth, and separated over S .*

Indeed, when H is smooth over S , so is $H \times_S G$, and one may apply 4.1 to the latter. By considering graph subgroups, $\text{Hom}_{S\text{-gr}}(H, G)$ becomes a subfunctor F' of the functor F of multiplicative-type subgroups of $H \times_S G$: namely, $F'(T)$ is the set of multiplicative-type subgroups K of

$$(H \times_S G)_T = H_T \times_T G_T$$

such that the homomorphism induced by the first projection,

$$K \longrightarrow H_T,$$

is an isomorphism. By Exposé IX, 2.9, the canonical morphism $F' \rightarrow F$ is an open immersion. Since F is representable, the same is true of F' , which is represented by an open subprescheme of F ; and since F is separated and smooth over S , so is F' .

When H is finite over S , it is enough to apply 3.12(b) to obtain the representability of $\text{Hom}_{S\text{-gr}}(H, G)$. When $H = H_1 \times_S H_2$, with H_1 smooth over S and H_2 finite over S , then $\text{Hom}_{T\text{-gr}}(H_T, G_T)$ is identified with the subset of

$$\text{Hom}_{T\text{-gr}}((H_1)_T, G_T) \times \text{Hom}_{T\text{-gr}}((H_2)_T, G_T)$$

formed by the pairs (u_1, u_2) such that $u_1 \cdot u_2 = u_2 \cdot u_1$. It follows that $\text{Hom}_{S\text{-gr}}(H, G)$ is represented by a closed subprescheme of

$$\text{Hom}_{S\text{-gr}}(H_1, G) \times_S \text{Hom}_{S\text{-gr}}(H_2, G) = X.$$

Indeed, apply Exposé VIII, 6.5(b), with $Y = H$, $Z = G$, and with q_1 and q_2 defined respectively by

$$(u_1, u_2) \longmapsto u_1 \cdot u_2 \quad \text{and} \quad (u_1, u_2) \longmapsto u_2 \cdot u_1.$$

In the general case, the question is local on S for the Zariski topology, so we may suppose that S is affine and that H has constant type over S . Since H is quasi-isotrivial (Exposé X, 4.5), there is a surjective étale morphism $S' \rightarrow S$, with S' affine, that splits H , that is, such that $H' = H_{S'}$ is diagonalizable. The preceding result then applies, because a diagonalizable group is the product of a diagonalizable torus and a finite diagonalizable group. It remains to see that the descent datum obtained on the S' -prescheme

$$\text{Hom}_{S'\text{-gr}}(H', G') = X'$$

is effective. This follows from the argument of Exposé X, 5.4, noting that, there, the hypothesis that $X' \rightarrow S'$ be separated and locally quasi-finite was used only to ensure that every open part of X' quasi-compact over S' was quasi-affine over S' . That property still holds here, as follows readily from the fact that it holds for the functor F of 4.1 (as will be seen during the proof of 4.1). This argument—or any similar variant of this small dévissage—establishes the representability of $\mathrm{Hom}_{S\text{-gr}}(H, G)$, and at the same time shows that it is separated over S . It is locally of finite presentation over S , as follows, for example, from the fact already noted in 3.2 that this functor “commutes with filtered direct limits of rings.” Finally, the functor is formally smooth over S by 2.1, and is therefore smooth over S .

Remark 4.3. — We have deduced 4.2 from 4.1; this is genuinely immediate only when H is also smooth over S . To make the deduction without contortions in the general case, one would need the representability assertion in 4.1 without assuming that G is smooth over S , assuming only that it is affine and of finite presentation over S . Of course, F would then no longer be smooth over S in general. There is little doubt that 4.1 remains true under these more general hypotheses, but the proof appears likely to be more delicate, because one can no longer invoke 3.8.¹²¹³

Let us nevertheless point out that when G is a closed subgroup of an affine smooth group G_0 over S , the functor F of multiplicative-type subgroups of G is represented by a closed subprescheme of the prescheme representing the analogous functor F_0 for G_0 , which is covered by 4.1; this follows readily by applying Exposé VIII, 6.4. This also raises the question whether an affine S -group scheme G , affine and of finite presentation over S , is isomorphic to a subgroup scheme of $\mathrm{GL}(n)_S$ for some suitable n . This is true when S is the spectrum of a field (cf. Exposé VI_B, 11.11), but is unfortunately false in general, even for tori; see 4.6. Finally, one could also prove 4.2 directly by exactly the same method as 4.1.

We now prove 4.1. Since the functor F is evidently local in nature, we may suppose that S is affine, say $S = \mathrm{Spec}(A)$, where A is a ring. Regarding A as the filtered direct limit of its subrings of finite type over $\mathbb{Z}$, and noting that G descends from a smooth affine group over one such subring (EGA IV, 8), we are reduced to the case in which S is noetherian.

For every integer $n > 0$, let T_n be the functor defined like F , but restricted to multiplicative-type subgroups H of G_T such that $n \cdot \mathrm{id}_H = 0$, that is, such that ${}_nH = H$. Order the set I of positive integers by divisibility. When m is a multiple of n , define

$$p_{n,m}: T_m \longrightarrow T_n$$

by

$$p_{n,m}(H) = {}_nH = \mathrm{Ker}(n \cdot \mathrm{id}_H).$$

In this way the T_n form an inverse system of functors over S . By 3.12(a), the functors T_n are representable, affine, and of finite type over S . Similarly define morphisms

$$u_n: F \longrightarrow T_n$$

¹²This is indeed proved for G flat and quasi-affine over S , with connected fibres, provided one restricts to the central subtori of G (Exposé XV, 8.8).

¹³Editor’s note: when H is smooth (not necessarily affine) over a normal locally noetherian base S , M. Raynaud showed that the largest representable open subfunctor of the functor of subtori of H is a disjoint sum of opens smooth and affine over S . This is Theorem IX.9.26 in *Faisceaux amples sur les schémas en groupes et sur les espaces homogènes*, Lecture Notes in Mathematics 119 (1970).

by $u_n(H) = {}_nH$. Thus one obtains a morphism

$$u: F \longrightarrow T = \varprojlim_n T_n,$$

where the inverse limit is taken in the category of functors over S . Since the T_n are representable and affine over S , so is T : it is the spectrum of the filtered direct limit of the quasi-coherent algebras on S defining the T_n . Of course, T is not in general of finite type over S .

We shall apply 3.7. It is enough to verify conditions (a)–(d) there; these will imply that F is represented by a prescheme locally of finite type over S . It then follows from 2.1 bis that F is even smooth over S . Moreover, since F is a subfunctor of T , which is affine over S , F is separated over S : it is separated over T , and T is separated over S . We can also prove at once the supplementary assertion used above: every open $U \subset F$ that is quasi-compact over S is quasi-affine over S . This follows from 3.11 and the fact that the T_n are affine over S .

Let us now verify the conditions of 3.7.

(a) F is a sheaf for the faithfully flat quasi-compact topology, by the descent theory of SGA 1, Exposé VIII, which applies here because groups of multiplicative type over T are affine over T .¹⁴ It commutes with filtered direct limits of rings by the standard machinery of EGA IV, 8. Let us show that it commutes with adic inverse limits of local artinian rings. For the analogous functor considered in 4.2, this is precisely Exposé IX, 7.1, in the special case where A is a complete noetherian local ring endowed with an ideal of definition for its usual topology. Notice that this is exactly where the hypothesis that G be affine enters essentially. In the present case we are reduced to the following lemma.

Lemma 4.4. — *Let A be a complete noetherian local ring with maximal ideal $\mathfrak{m}$, let G be a group scheme affine over $S = \operatorname{Spec}(A)$, and, for every integer $n \geq 0$, put*

$$S_n = \operatorname{Spec}(A/\mathfrak{m}^{n+1}), \quad G_n = G \times_S S_n.$$

For every n , let H_n be a multiplicative-type subgroup of finite type of G_n , and suppose that, whenever $m \geq n$, H_n is obtained from H_m by reduction. Then there exists a unique multiplicative-type subgroup H of G whose reductions are the H_n .

By Exposé X, 3.2, there exists a multiplicative-type group H over S , necessarily of finite type and isotrivial and determined up to unique isomorphism, together with an isomorphism

$$H \times_S S_0 \simeq H_0.$$

Here one uses the fact that H_0 is isotrivial, since it is of finite type over a field; see Exposé X, 1.4. By Exposé X, 2.1, for every n the isomorphism $H \times_S S_0 \simeq H_0$ lifts to a unique isomorphism $H \times_S S_n \simeq H_n$. Exposé IX, 7.1, already cited, now shows that the homomorphisms $H_n \rightarrow G_n$ arise from a unique homomorphism of S -groups

$$u: H \longrightarrow G.$$

By Exposé IX, 6.6, this homomorphism is a monomorphism because $u_0: H_0 \rightarrow G_0$ is one. This proves 4.4.

(b) The morphism $u: F \rightarrow T$ is a monomorphism. This follows from the density theorem, Exposé IX, 4.7, in the form of Corollary 4.8(b). It is essential for the present application

¹⁴The printed source says “affine over S ” here; the descent assertion concerns groups over T , so the required base is T .

that this result is available over a base that need not be noetherian. Since the functor T does not commute with filtered direct limits of rings, one cannot reduce to the noetherian case a priori.

(b') The morphism $u: F \rightarrow T$ is infinitesimally étale; in other words, we have the following lemma.

Lemma 4.5. — *Let A be an artinian local ring with residue field k , let $S = \operatorname{Spec}(A)$, let $I \subset \operatorname{rad}(A)$ be an ideal, and put $S_0 = \operatorname{Spec}(A/I)$. Let G be an S -group prescheme smooth over S , and put $G_0 = G \times_S S_0$. Suppose given a multiplicative-type subgroup H_0 of G_0 and, for every integer $n > 0$, a multiplicative-type subgroup $H(n)$ of G , in such a way that*

1° *for every multiple m of n , $H(n) = {}_nH(m)$; and*

2° *$H(n)_0 = {}_nH_0$.*

Then there exists a unique multiplicative-type subgroup H of G such that ${}_nH = H(n)$ for every n .

Uniqueness is already contained in (b). For existence, an immediate induction reduces us to the case in which $I\mathfrak{m} = 0$, where $\mathfrak{m}$ is the maximal ideal of A .¹⁵ Put

$$k = A/\mathfrak{m}, \quad S_0 = \operatorname{Spec}(k), \quad G_0 = G \times_S S_0,$$

and let $\mathfrak{g}_0$ and $\mathfrak{h}_0$ be the Lie algebras of G_0 and H_0 , respectively. There is a canonical isomorphism of groups

$$\mathfrak{g}_0 \otimes_k I \simeq \operatorname{Ker}(G(S) \rightarrow G(S_0));$$

see Exposé III. By Exposé IX, 3.6 bis and 3.7, there exists a multiplicative-type subgroup H of G reducing to H_0 , and such an H is determined, up to an inner automorphism by an element of

$$N = \operatorname{Ker}(G(S) \rightarrow G(S_0)).$$

Thus the set P of liftings H of H_0 is a principal homogeneous set under N/M , where

$$M = \{g \in N \mid \operatorname{int}(g) \cdot H = H\}.$$

One sees readily (Exposé III) that this subgroup is precisely the vector subspace of $\mathfrak{g}_0 \otimes_k I$ consisting of the invariants under H_0 , where H_0 acts on $\mathfrak{g}_0 \otimes_k I$ through the representation induced by the adjoint representation of G_0 . Likewise, the set $P(n)$ of liftings of $H(n)_0 = {}_nH_0$ is a principal homogeneous set under $N/M(n)$, where $M(n)$ is the vector subspace of $\mathfrak{g}_0 \otimes_k I = N$ consisting of the invariants under ${}_nH_0$. The density theorem, Exposé IX, 4.7, shows readily that $M = M(n)$ for sufficiently large n , where “large” is understood with respect to the divisibility order on the positive integers. Consequently, the natural map

$$H \mapsto {}_nH$$

from P to $P(n)$, which is compatible with the actions of N , and hence with the homomorphism

$$N/M \rightarrow N/M(n)$$

between the acting groups, is bijective for sufficiently large n . The conclusion of 4.5 follows immediately.

¹⁵The printed source abruptly writes $J\mathfrak{m} = 0$ here and uses J throughout the ensuing argument, although only I was introduced in the statement. We retain I , the same ideal, throughout.

To verify conditions (c) and (d) of 3.7, we use 3.8, which reduces us to checking (c') and (d') below.

(c') The T_n are smooth over S , and the transition morphisms $p_{n,m}: T_m \rightarrow T_n$ are smooth.

This is precisely 2.2 bis.

(d') With the notation of 3.8, a point ξ of F with values in a field k over S is just a multiplicative-type subgroup H_0 of $G_0 = G_k$. Repeating the argument in (b') above, we see that the integer $d(\xi)$ considered in 3.8 is

$$d(\xi) = \dim_k(\mathfrak{g}_0/\mathfrak{g}_0^{H_0}),$$

where $\mathfrak{g}_0$ is the Lie algebra of G_0 and $\mathfrak{g}_0^{H_0}$ is the vector subspace of invariants under H_0 . This is a finite integer, so condition (d')^{1°} of 3.8 holds.

With the notation of (d')^{2°}, giving a morphism $v: X \rightarrow F$ amounts to giving a multiplicative-type subgroup H of G_X . For $x \in X$, the integer $d(\xi_x)$ ¹⁶ is then the dimension of

$$(\mathfrak{g} \otimes \kappa(x))/(\mathfrak{g} \otimes \kappa(x))^{H_x}.$$

where $\mathfrak{g}$ is the sheaf of Lie algebras of G_X . It is locally free of finite type over X , because G_X is smooth over X , and $\mathfrak{g} \otimes \kappa(x)$ is the Lie algebra of the fibre G_x of G_X at x ; here H_x is the fibre of H at x . Now $\mathfrak{g}$ is a locally free module on X ¹⁷ on which the multiplicative-type group H acts, and the invariant subfunctor $\mathfrak{g}^H$ is therefore given by a locally direct-summand, hence locally free, subsheaf of $\mathfrak{g}$. By descent, one reduces to the case in which H is diagonalizable, and then applies Exposé I, 4.7.3, noting that the invariant subsheaf corresponds to the degree-zero component. Consequently,

$$d(\xi_x) = \text{rank}_x(\mathfrak{g}/\mathfrak{g}^H),$$

so this is a locally constant function of x . This proves condition (d').

We have thus verified the conditions of 3.7, completing the proof of 4.1.

Remark 4.6. — When $G = \text{GL}(n)_S$, one can give a much simpler and more explicit direct proof of 4.1, using Exposé I, 4.7.3. The proof moreover shows that, in this case, the moduli scheme is a disjoint sum of affine S -schemes. Proceeding as indicated in 4.3, one obtains the same result whenever G is a closed subgroup of a group of the form $\text{GL}(n)_S$.

One must not, however, suppose that the preschemes representing the functors in 4.1 or 4.2 are always disjoint sums of affine S -schemes. For example, let H be a multiplicative-type group of finite type over a locally noetherian prescheme S . By Exposé X, 5.11, H is isotrivial if and only if

$$\text{Hom}_{S\text{-gr}}(H, \mathbb{G}_m)$$

is a disjoint sum of preschemes affine over S . But, as noted in Exposé X, 1.6, there may exist nonisotrivial tori H of relative dimension 2. For such a torus, $\text{Hom}_{S\text{-gr}}(H, \mathbb{G}_m)$ is not a disjoint sum of S -preschemes affine over S , and likewise the “dual” twisted constant group

$$\text{Hom}_{S\text{-gr}}(\mathbb{G}_m, H)$$

¹⁶The printed source gives only $(\mathfrak{g} \otimes \kappa(x))^{H_x}$ here and later $\text{rank}_x \mathfrak{g}^H$, omitting the quotient. That reading contradicts both the immediately preceding formula $d(\xi) = \dim(\mathfrak{g}_0/\mathfrak{g}_0^{H_0})$ and the lifting torsor N/M . The quotient is restored in both places.

¹⁷The printed source says “over S ” here, although $\mathfrak{g}$ is the Lie-algebra sheaf of G_X ; the base is therefore X .

is not such a sum. Indeed, if two commutative twisted constant groups R, R' of finite presentation are dual to one another,

$$R' = \mathrm{Hom}_{S\text{-gr}}(R, \mathbb{Z}_S),$$

then one readily sees that one is isotrivial if and only if the other is.

This last point also shows that such an H is not isomorphic to a subgroup of a group of the form $\mathrm{GL}(n)_S$. More precisely, one can show that a multiplicative-type group of finite type over a connected locally noetherian S is isomorphic to a subgroup of a group of the form $\mathrm{Aut}_{\mathrm{Mod}}(E)$, where E is a locally free module of finite type over S , if and only if it is isotrivial. Finally, taking $G = H \times \mathbb{G}_m$ in the two preceding examples gives an example in which the prescheme representing the functor F of 4.1 is not a disjoint sum of affine S -preschemes; if desired, G may be taken to be a torus of relative dimension 3.

5. First Corollaries of the Representability Theorem

Let H and G be as in 4.2. Put

$$M = \mathrm{Hom}_{S\text{-gr}}(H, G),$$

which, by 4.2, is a smooth S -prescheme separated over S . Notice that G acts on the functor $M = \mathrm{Hom}_{S\text{-gr}}(H, G)$ by

$$(g, u) \longmapsto \mathrm{int}(g) \circ u.$$

Hence there is a canonical morphism

$$(x) \quad G \times_S M \longrightarrow M \times_S M,$$

whose components are the preceding morphism $G \times_S M \rightarrow M$ and the second projection $G \times_S M \rightarrow M$.

Corollary 5.1. — *The preceding morphism (x) is smooth.*

This follows from 4.2 and 2.3. This statement is equivalent to the following one.

Corollary 5.2. — *Let $u_1, u_2 : H \rightrightarrows G$ be two homomorphisms of S -groups. Then the subfunctor $\mathrm{Transp}(u_1, u_2)$ of G (cf. 2.4) is representable by a closed subprescheme of G , smooth over S .*

It remains only to prove that $\mathrm{Transp}(u_1, u_2) \rightarrow G$ is indeed a closed immersion. This follows because it is the equalizer of a pair of morphisms $G \rightrightarrows M$, namely

$$g \longmapsto \mathrm{int}(g)u_1 \quad \text{and} \quad g \longmapsto u_2,$$

the latter being the “constant” morphism, and because M is separated over S . In particular:

Corollary 5.3. — *Let $u : H \rightarrow G$ be a morphism of S -groups. Then*

$$\mathrm{Centr}_G(u) = \mathrm{Transp}(u, u)$$

is representable by a closed group subprescheme of G , smooth over S . Moreover, the quotient of G by $\mathrm{Centr}_G(u)$ is representable by an open subprescheme of M .

It remains to prove the last assertion. The morphism

$$g \longmapsto \mathrm{int}(g) \circ u$$

from G to M is smooth and of finite type by 5.2, hence is an open morphism (EGA IV, 6.6). Let U denote its image, equipped with the structure induced from M . The induced morphism $G \rightarrow U$ is smooth, surjective, and of finite type, and is therefore a covering for the faithfully flat quasi-compact topology. Moreover, it is clear that the preceding morphism $G \rightarrow M$ makes G a formally principal homogeneous sheaf under $\text{Centr}_G(u)_M$. It follows that the sheaf $G/\text{Centr}_G(u)$ is represented by U .

Corollary 5.4. — *Let $u_1, u_2 : H \rightrightarrows G$ be two homomorphisms of S -groups. The following conditions are equivalent:*

- (i) u_1 and u_2 are conjugate locally for the étale topology.
- (i bis) u_1 and u_2 are conjugate locally for the faithfully flat quasi-compact topology.
- (ii) For every $s \in S$, let $\bar{s}$ denote the spectrum of an algebraic closure of $\kappa(s)$. The morphisms

$$u_{1,\bar{s}}, u_{2,\bar{s}} : H_{\bar{s}} \rightrightarrows G_{\bar{s}}$$
 are conjugate by an element of $G(\kappa(\bar{s}))$.
- (ii bis) The structural morphism $\text{Transp}(u_1, u_2) \rightarrow S$ is surjective.
- (iii) $\text{Transp}(u_1, u_2)$ is a torsor under the action of the smooth S -group prescheme of finite type $\text{Centr}_G(u_1)$.

The implications (i) $\Rightarrow$ (i bis) and (ii) $\Rightarrow$ (ii bis) are immediate, the latter by the Nullstellensatz. The implication (i bis) $\Rightarrow$ (ii) follows from the “principle of finite extension” (EGA IV, 9). Moreover, (ii bis) $\Rightarrow$ (iii): since $\text{Transp}(u_1, u_2)$ is smooth over S , it is flat over S , and since it is of finite type, it is quasi-compact over S . Thus it is faithfully flat and quasi-compact over S if and only if its structural morphism is surjective. On the other hand, it is formally principal homogeneous under $\text{Centr}_G(u_1)$, which is faithfully flat and quasi-compact over S . The last condition is therefore also equivalent to saying that $\text{Transp}(u_1, u_2)$ is a torsor under $\text{Centr}_G(u_1)$, in the faithfully flat quasi-compact topology. Finally, (iii) $\Rightarrow$ (i) follows from the smoothness of $\text{Transp}(u_1, u_2)$ over S and from Hensel’s lemma in the form 1.10.

Remark 5.5. — Fix $u_1 = u : H \rightarrow G$. Consider the functor

$$(\text{Sch})^\circ/S \longrightarrow (\text{Ens})$$

that assigns to every T over S the set of homomorphisms of T -groups $u_2 : H_T \rightarrow G_T$ that are locally conjugate, for the étale topology, to $u_T : H_T \rightarrow G_T$. This functor is represented precisely by the open subscheme of M , isomorphic to $G/\text{Centr}_G(u)$, considered in 5.3.

Let us sketch the variants of the preceding results obtained by applying 4.1 instead of 4.2. Thus let G be a group prescheme smooth and affine over S , and now let M denote the smooth S -prescheme separated over S that represents the functor considered in 4.1. Again G acts on M :

$$(g, H) \longmapsto \text{int}(g)(H).$$

Hence, as above, there is a morphism

$$(x \text{ bis}) \quad G \times_S M \longrightarrow M \times_S M.$$

Corollary 5.1 bis. — *The preceding morphism is smooth.*

This follows from 4.1 and 2.3 bis. We again conclude:

Corollary 5.2 bis. — *Let H_1, H_2 be two multiplicative-type subgroups of G . Then the subfunctor $\mathrm{Transp}_G(H_1, H_2)$ of G is representable by a closed subprescheme of G , smooth over S .*

In particular:

Corollary 5.3 bis. — *Let H be a multiplicative-type subgroup of G . Then the group subfunctor $\mathrm{Norm}_G(H)$ of G is representable by a closed subprescheme of G , smooth over S . Moreover, the quotient $G/\mathrm{Norm}_G(H)$ is representable by an open subprescheme of M .*

Corollary 5.4 bis. — *Let H_1, H_2 be two multiplicative-type subgroups of G . The following conditions are equivalent:*

- (i) H_1 and H_2 are conjugate locally for the étale topology.
- (i bis) H_1 and H_2 are conjugate locally for the faithfully flat quasi-compact topology.
- (ii) For every $s \in S$, let $\bar{s}$ denote the spectrum of an algebraic closure of $\kappa(s)$. The subgroups $H_{1, \bar{s}}$ and $H_{2, \bar{s}}$ of $G_{\bar{s}}$ are conjugate by an element of $G(\kappa(\bar{s}))$.
- (ii bis) The structural morphism $\mathrm{Transp}(H_1, H_2) \rightarrow S$ is surjective.
- (iii) $\mathrm{Transp}(H_1, H_2)$ is a principal homogeneous bundle under the action of the smooth S -group prescheme of finite type $\mathrm{Norm}_G(H_1)$.

Remark 5.5 bis. — Remark 5.5 carries over to the present case as well.

Remark 5.6. — To prove 5.2, and hence also the first assertion of 5.3 and the assertions of 5.4, the reference to 4.2 may be replaced by a reference to VIII, 6.4, whose proof is much easier. This also shows that the hypothesis that G be affine over S is unnecessary there. Moreover, in Section 6 we shall show how a variant of this method extends these results to certain groups H more general than groups of multiplicative type. The same observations apply to the variants 5.2 bis and following. By contrast, the result 5.8 below makes essential use of all the hypotheses imposed—in particular, that G be affine and smooth over S , and that H be of multiplicative type—and the lecturer knows no proof other than one using the representability theorems 4.1 or 4.2.¹⁸

5.7. Since the morphism (x), respectively (x bis), is smooth and hence open, its image is open. Let R be this image, equipped with the structure induced from $M \times_S M$. One checks readily that R is an equivalence relation on M , namely the relation made explicit in 5.4, respectively 5.4 bis. It would be interesting to know whether the quotient sheaf M/R , which is formally étale over S , is representable; if so, it is represented by a prescheme étale over S . This is the case, for example, when S is the spectrum of a field. Notice also that R need not be closed in $M \times_S M$, even when G is quasi-finite over S . Thus, when M/R is representable, it may be nonseparated over S .

Theorem 5.8. — *Let S be a prescheme, let G be a smooth affine S -prescheme, and let $s \in S$. Then:*

- (a) *For every multiplicative-type subgroup H_0 of G_s , there exist an étale morphism $S' \rightarrow S$, a point $s' \in S'$ lying over s such that the residue-field extension $\kappa(s')/\kappa(s)$ is trivial, and a multiplicative-type subgroup H' of $G' = G \times_S S'$, such that*

$$H'_{s'} = H_0 \otimes_{\kappa(s)} \kappa(s').$$

¹⁸The situation has changed since this text was written; cf. Exposés XV and XIX, no. 6.

- (b) For every group homomorphism $u_0 : H_s \rightarrow G_s$, where H is a multiplicative-type S -group prescheme of finite type, there exist an étale morphism $S' \rightarrow S$, a point $s' \in S'$ lying over s such that $\kappa(s')/\kappa(s)$ is trivial, and a group homomorphism

$$u' : H' \longrightarrow G', \quad H' = H \times_S S', \quad G' = G \times_S S',$$

such that

$$u'_{s'} : H'_{s'} \longrightarrow G'_{s'} \quad \text{equals} \quad u_0 \otimes_{\kappa(s)} \kappa(s').$$

This follows from 4.1, respectively 4.2, and from Hensel's lemma in the form 1.10.

Proposition 5.9. — *Let S be a prescheme, let G be a smooth affine S -prescheme, and let H be a multiplicative-type subgroup of G . Consider*

$$\text{Centr}_G(H) \hookrightarrow \text{Norm}_G(H),$$

which, by 5.3 and 5.3 bis, are closed group subpreschemes of G , smooth over S . The first is then an open and closed subprescheme of the second, and the quotient sheaf

$$W_G(H) = \text{Norm}_G(H) / \text{Centr}_G(H)$$

is represented by an open group subprescheme of $\text{Aut}_{S\text{-gr}}(H)$. It is therefore a quasi-finite, étale, and separated S -group prescheme.

Indeed, consider the evident homomorphism

$$\theta : \text{Norm}_G(H) \longrightarrow \text{Aut}_{S\text{-gr}}(H),$$

whose kernel is, by definition, $\text{Centr}_G(H)$. Since $\text{Aut}_{S\text{-gr}}(H)$ is represented by an étale S -group prescheme separated over S (X, 5.10), its unit section is an open and closed immersion. Its inverse image under θ is therefore an open and closed subgroup of $\text{Norm}_G(H)$.

Moreover, θ is smooth. This follows formally from the definitions and from the facts that $\text{Norm}_G(H)$ is smooth over S and that $\text{Aut}_{S\text{-gr}}(H)$ is étale over S . As in 5.3, the image of θ is consequently an open subprescheme U of $\text{Aut}_{S\text{-gr}}(H)$, and, with the induced structure, U represents the quotient sheaf $\text{Norm}_G(H) / \text{Centr}_G(H)$. The latter is therefore étale and separated over S . It is also quasi-finite over S , since it is quasi-compact over S , being the image of $\text{Norm}_G(H)$, which is quasi-compact. This proves 5.9.

Corollary 5.10. — *For every $s \in S$, put*

$$w(s) = \text{rank}(\text{Norm}_{G_s}(H_s) / \text{Centr}_{G_s}(H_s)).$$

This is also the index of $\text{Centr}_{G(\bar{k})}(H(\bar{k}))$ in $\text{Norm}_{G(\bar{k})}(H(\bar{k}))$, where $\bar{k}$ is an algebraic closure of $\kappa(s)$. Then the function $s \mapsto w(s)$ is lower semicontinuous. It is constant in a neighborhood of s if and only if $W_G(H)$ is finite over S in a neighborhood of s .

Indeed, for every S -prescheme W that is étale, of finite type, and separated over S , the function

$$s \longmapsto \text{rank}[W_s : \kappa(s)]$$

is lower semicontinuous, and it is constant in a neighborhood of s if and only if W is finite over S in a neighborhood of s . This fact was noted in SGA 1, I, 10.9; its proof, which presents no difficulty, is given in EGA IV.¹⁹

¹⁹EGA IV, 15.5.1 and 18.10.7.

Remark 5.11.²⁰ — Let G be a group prescheme affine and smooth over S , and let H be a multiplicative-type subgroup. Then 5.3 and 5.3 bis imply that the quotients

$$G/\mathrm{Centr}_G(H) \quad \text{and} \quad G/\mathrm{Norm}_G(H)$$

are preschemes smooth and quasi-affine over S . Indeed, in both cases the moduli prescheme M in which the quotient is embedded has the property that every open $U \subset M$ quasi-compact over S is quasi-affine over S , by 3.11.

We shall moreover see in the next Exposé²¹ that for every such U , the schematic closure $\overline{U}$ of U in M is even affine over S . Thus the quotients under consideration are affine over S , provided that the open subprescheme U of the corresponding moduli prescheme—of group homomorphisms from H to G , respectively of multiplicative-type subgroups of G —is closed in M . This is the case, for example, when H is a “maximal torus” in the second case, or when S is the spectrum of a field; cf. XII.5.4.²² I do not know whether the quotients under consideration are affine over S in general.

6. A Rigidity Property for Homomorphisms of Certain Group Schemes, and the Representability of Certain Transporters²³

Proposition 6.1. — *Let S be a locally noetherian prescheme, let H be a commutative S -group prescheme of finite type over S , and let E be a set of positive integers stable under multiplication. Suppose that the following conditions hold:*

(a) *for every $n \in E$, the subgroup*

$${}_nH = \mathrm{Ker}(n \cdot \mathrm{id}_H)$$

is finite and flat over S ;

(b) *for every $s \in S$, the family of the ${}_nH_s$, $n \in E$, is schematically dense in the fibre H_s .*²⁴

*Let Y be a closed subprescheme of H . Then the subfunctor $\prod_{H/S}(Y/H)$ of S (compare Exposé VIII, 6) is representable by a closed subprescheme T of S . If S is noetherian, there exists $n \in E$ such that*²⁵

$$(x) \quad \prod_{H/S}(Y/H) = \prod_{{}_nH/S} ((Y \cap {}_nH)/{}_nH).$$

²⁰Editor’s note: For a generalization to the nonaffine case, see M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes*, Lecture Notes in Mathematics 119 (1970), IX.2.8.

²¹Editor’s note: Cf. Remark XII.5.7. We mention here the following generalization. Without assuming G affine, if H is a torus and S is locally noetherian, U is affine and smooth by a theorem of Raynaud (*loc. cit.*, IX.2.6 and IX.2.8). Indeed, $G/\mathrm{Norm}_G(H)$ is an open subprescheme of M , and the largest representable open subfunctor of M is a disjoint sum of open subpreschemes smooth and affine over S .

²²Editor’s note: This assertion assumes that G has locally constant reductive rank.

²³The present section does not use the results of Sections 3, 4, and 5; its natural place would be in Exposé VI_B.

²⁴The printed French text has H here; the fibrewise condition and its use in the proof require H_s .

²⁵The printed formula writes the right-hand quotient as $(Y \cap {}_nH)/S$. Since the product functor is taken over ${}_nH/S$, and as the proof requires, the quotient is $(Y \cap {}_nH)/{}_nH$.

Indeed, by Exposé VIII, 6.4, condition (a) implies that ${}_nH$ is finite and flat, and hence a fortiori “essentially free” over S . Thus the right-hand side of (x) is representable by a closed subprescheme T_n of S .²⁶ Order E by divisibility. The T_n then form a decreasing family of closed subpreschemes of S ; consequently, when S is noetherian—as we may now suppose—this family is stationary for sufficiently large n . Let T be the stable value of T_n . We claim that T represents the left-hand side of (x), which will complete the proof.

It is enough to prove the following. Let S' be a prescheme over S such that

$$(Y \cap {}_nH)_{S'} = ({}_nH)_{S'} \quad \text{for every } n \in E,$$

or, equivalently, such that $Y_{S'} \supset ({}_nH)_{S'}$ for every $n \in E$. Then $Y_{S'} = H_{S'}$. This follows at once from Exposé IX, 4.4: in view of conditions (a) and (b), the family of the $({}_nH)_{S'}$ is schematically dense in $H_{S'}$.

Theorem 6.2. — *Let S be a prescheme, let H, E be as in 6.1 and satisfy conditions (a) and (b), let $u : H \rightarrow G$ be a homomorphism of S -groups from H to an S -group prescheme locally of finite type over S , and let K be a closed subgroup prescheme of G .²⁷ Consider the subfunctor $\text{Transp}_G(u, K)$ of G (cf. 2.5). It is representable by a closed subprescheme of G . If G is noetherian—for example, if S is noetherian and G is of finite type over S —there exists $n \in E$ such that*

$$\text{Transp}_G(u, K) = \text{Transp}_G(u|_{{}_nH}, K).$$

Suppose, finally, that G is smooth over S , that K is smooth over S or is of multiplicative type, and that H, E satisfy the following condition, stronger than (a):

a' for every $n \in E$, the subgroup ${}_nH$ of H is of multiplicative type.

Then $\text{Transp}_G(u, K)$ is smooth over S .

Consider the G -group

$$H_G = H \times_S G,$$

which plainly satisfies the conditions of 6.1, with S replaced by G and H by H_G . Let Y be the closed subprescheme of H_G that is the inverse image of $K \subset G$ under the map

$$H_G = H \times_S G \longrightarrow G, \quad (h, g) \longmapsto \text{int}(g) \cdot u(h).$$

Then

$$\text{Transp}_G(u, K) = \prod_{H_G/G} (Y/H_G)$$

(compare Exposé VIII, Example 6.5(e)). The first assertions therefore follow from 6.1. Moreover, for every quasi-compact open $U \subset G$, there exists $n \in E$ for which the restrictions to U of the two functors

$$\text{Transp}_G(u, K) \quad \text{and} \quad \text{Transp}_G(u|_{{}_nH}, K)$$

coincide. To verify the last assertion of 6.2, we may therefore replace H by some ${}_nH$, after which it is enough to apply 2.5, since ${}_nH$ is assumed to be of multiplicative type over S .

Remark 6.3. — The preceding proof uses only the very elementary result of Exposé VIII, 6.4, together, for its last part, with 2.5; when K is smooth over S , the latter is

²⁶The printed French text has “ T_n of T ” here; the next sentence makes the T_n closed subpreschemes of S , and T is defined only afterward as their stable value.

²⁷The printed French text has H here; the proof immediately uses $K \subset G$, as required by the transporter.

the infinitesimal result of Exposé IX, 3.6, and hence the vanishing of the cohomology of multiplicative-type groups. In the most important cases (cf. 6.7), the integers $n \in E$ may even be taken prime to the residue characteristics of S , that is, invertible in $\mathcal{O}_S$. The ${}_nH$ are then finite étale over S , and the cohomological result invoked is practically trivial. Thus, in that situation, 6.2 is independent of the theory of multiplicative-type groups.

6.4. — As usual, one sees that 6.1 extends to the following situation. Let S' be an S -prescheme (not necessarily locally noetherian), put $H' = H_{S'}$, and let Y' be a closed subprescheme of H' , provided that $Y' \rightarrow H'$ is of finite presentation, that is, that the ideal defining Y' is of finite type. Then $\prod_{H'/S'}(Y'/H')$ is representable by a closed subprescheme T' of S' , the morphism $T' \rightarrow S'$ is of finite presentation, and, if S' is quasi-compact, there exists $n \in E$ for which the analogue of (x) holds. It follows as well that the first assertion of 6.2 remains valid without assuming that G is locally of finite type over S , provided that the immersion $K \rightarrow G$ is of finite presentation.

6.5. — As announced in 5.6, Theorem 6.2 makes it possible to extend to group preschemes satisfying (a') and (b) above certain results proved by another method, under more restrictive hypotheses, for groups of multiplicative type. This applies to 5.2, the beginning of 5.3, and 5.4 and their corresponding “bis” variants, and also to 5.9 and 5.10. It also applies to the results of Exposé IX, Sections 5 and 6, with the exception of Exposé IX, 6.8. The latter assertion is already false for a homomorphism of abelian schemes $u : H \rightarrow G$ over the spectrum S of a discrete valuation ring of residue characteristic $p > 0$: it can happen that $\text{Ker } u$ has the trivial group as its generic fibre and, as its special fibre, a radicial group that is not reduced to the trivial group.

6.6. — We have just given an example of a rigidity property of multiplicative-type groups that is not shared by abelian schemes. Another, extremely important, example is that the existence theorem for infinitesimal extensions of homomorphisms in Exposé IX, 3.6 no longer holds when H is an abelian scheme. Thus a nonzero abelian scheme over a field admits nontrivial infinitesimal variations, in contrast to a multiplicative-type group. This is the infinitesimal aspect of the fact that there is a theory of moduli—one that, moreover, is far from complete—for abelian varieties, whereas the theory of moduli of multiplicative-type groups is empty.

Another, “global,” aspect of this infinitesimal difference is the following. Let H be an abelian scheme over a locally noetherian prescheme S , and let G be a commutative S -group prescheme locally of finite type over S . One can show that $\text{Hom}_{S\text{-gr}}(H, G)$ is representable by a prescheme locally of finite type over S . Unlike the corresponding prescheme when H is of multiplicative type, however, it is not étale over S , but only unramified. Thus, for example, if S is the spectrum of a complete discrete valuation ring and H, G are abelian schemes over S , there may exist homomorphisms $H_0 \rightarrow G_0$ on the special fibres that do not arise “by specialization” from a homomorphism on the generic fibres.

6.7. — Theorem 6.2 applies whenever H is an abelian scheme over S , or, more generally, an extension of such a scheme by a torus. Indeed, the question is local on S , so one may suppose that there is a prime number ℓ prime to the residue characteristics of S . Taking E to be the set of powers of ℓ then suffices to satisfy conditions (a') and (b).

An argument analogous to that of 6.1 gives the following result.

Theorem 6.8. — *Let X be a prescheme smooth over S , with nonempty connected geometric fibres. For every closed subprescheme Y of X , the functor $\prod_{X/S}(Y/X)$ is representable by a closed subprescheme T of S . If Y is of finite presentation over X , then T is of finite presentation over S .*

Since $f : X \rightarrow S$ is faithfully flat and locally of finite presentation, it is a covering for the fpqc topology. On the other hand, $T = \prod_{X/S}(Y/X)$ is plainly a subsheaf of S for the fpqc topology. Hence both the representability of T by a closed subscheme of S and the question whether T is of finite presentation over S are local on S for the fpqc topology. After the base change $S' \rightarrow S$ with $S' = X$, we are reduced to the case in which X admits a section e over S . We may also suppose that S is affine, and hence in particular quasi-compact. We then have:

Corollary 6.9. — *Under the hypotheses of 6.8, suppose that S is quasi-compact and that X admits a section e over S . For every integer $n \geq 0$, let X_n be the order- n infinitesimal neighbourhood of the section e in X . Suppose that Y is of finite presentation over X . Then there exists an integer $n \geq 0$ such that*

$$\prod_{X/S} (Y/X) = \prod_{X_n/S} (Y_n/X_n), \quad Y_n = Y \cap X_n.$$

This corollary implies 6.8 when Y is of finite presentation over X , by Exposé VIII, 6.4. Indeed, since X is smooth over S , X_n is finite and locally free over S , and a fortiori “essentially free” over S in the sense of Exposé VIII, 6.1. Thus $\prod_{X_n/S}(Y_n/X_n)$ is representable by a closed subscheme of S . Furthermore, either the proof of the cited result or reduction to the noetherian case shows at once that this closed subscheme is of finite presentation over S .

Let us first prove 6.9, and hence 6.8, when S is noetherian. Put

$$T_n = \prod_{X_n/S} (Y_n/X_n).$$

The T_n form a decreasing sequence of closed subschemes of S , and this sequence is stationary because S is noetherian. Let

$$R = \bigcap_{n \geq 0} \prod_{X_n/S} (Y_n/X_n)$$

be their common value for large n . Plainly $T \subset R$, and it remains to show that $R \subset T$. After the base change $R \rightarrow S$, we are reduced to the case $R = S$, that is, $Y_n = X_n$ for every n , or equivalently $Y \supset X_n$ for every n . We must then prove $T = S$, that is, $Y = X$.

The inclusions $Y \supset X_n$ for all n , together with the fact that X is locally noetherian, imply that in a neighbourhood of each point of $e(S)$, Y is an induced open subscheme of X . Hence there is an induced open subscheme U of X such that $e(S) \subset U \subset Y$. By Exposé IX, 4.3, since the fibres of X/S are integral, U is schematically dense in X . As Y is a closed subscheme containing U , it follows that $Y = X$. This proves 6.9, and hence 6.8, in the noetherian case.

The general case is reduced to the preceding one. For every $s \in S$, there is an affine open neighbourhood U of s and an affine open neighbourhood V of $e(s)$ such that $f(V) \subset U$. Then $f(V)$ is an open neighbourhood of s contained in U . Choose an affine open neighbourhood S' of s contained in $e^{-1}(V) \cap f(V)$, and put $X' = V \cap f^{-1}(S')$. Then X' and S' are affine open subschemes of X and S , respectively, and X'/S' admits a section. By the local nature of 6.8 and 6.9, we may suppose that $S = S'$.

We claim that

$$\prod_{X/S} (Y/X) = \prod_{X'/S'} (Y'/X'), \quad Y' = Y \cap X'.$$

Indeed, by Exposé IX, 4.6, X' is schematically dense in X , at least when X is quasi-separated over S , so that X' is retrocompact in X . In fact, one can readily prove that Exposé IX, 4.6 remains valid without the retrocompactness hypothesis. Likewise, after every base change $S_1 \rightarrow S$, $X' \times_S S_1$ is schematically dense in $X \times_S S_1$, and the claimed equality follows immediately. We are thus reduced to the case $X = X'$, so we may suppose that both S and X are affine.

Now let $X = \operatorname{Spec}(B)$, and let J be the ideal of B defining Y . The ideal J is the filtered colimit of its finite-type subideals. Consequently, Y is the intersection of closed subpreschemes Y_i of X that are of finite presentation over S , and

$$\prod_{X/S} (Y/X) = \bigcap_i \prod_{X/S} (Y_i/X).$$

To prove 6.8, it is therefore enough to treat the case in which Y is of finite presentation over X . It remains to prove 6.9 with S and X affine. But X and Y over S then arise by base change $S \rightarrow S_0$ from an analogous situation X_0, Y_0 over a noetherian prescheme S_0 . This reduces us to the already treated case in which S is noetherian, and completes the proof of 6.8 and 6.9.

Corollary 6.10. — *Let X be an S -group prescheme, smooth and of finite presentation, with connected fibres. Let Y be a group prescheme of finite presentation over S , and let $i : Y \rightarrow X$ be a monomorphism of S -group preschemes, so that Y is a subgroup of X . Then $\prod_{X/S} (Y/X)$ is representable by a closed subprescheme of S of finite presentation. If S is quasi-compact, let X_n denote, for every $n \geq 0$, the order- n infinitesimal neighbourhood of the unit section of X , and put $Y_n = X_n \cap Y$. Then, for all sufficiently large n ,*

$$\prod_{X/S} (Y/X) = \prod_{X_n/S} (Y_n/X_n).$$

The proof is essentially that of 6.9. The unit sections of X and Y induce bijective immersions $S \rightarrow X_n$ and $S \rightarrow Y_n$, and hence isomorphisms

$$S_{\text{red}} \xrightarrow{\sim} (X_n)_{\text{red}}, \quad S_{\text{red}} \xrightarrow{\sim} (Y_n)_{\text{red}}.$$

It follows that the injection $Y_n \rightarrow X_n$ is proper. Since it is a monomorphism of finite presentation, it is therefore a closed immersion. Exposé VIII, 6.4, already used above, now shows that $\prod_{X_n/S} (Y_n/X_n)$ is representable by a closed subprescheme of S of finite presentation over S . It remains to prove the last assertion of 6.10 under the additional assumption that S is affine.

We reduce once more to the case in which S is noetherian. With the notation of the proof of 6.9, it is enough to prove $R = T$, or, equivalently, that $Y \supset X_n$ for every n implies $Y = X$. This hypothesis implies that $i : Y \rightarrow X$ is étale at the points of the unit section of Y over S , and therefore that Y is smooth over S at those points. Hence the open subprescheme Y' consisting of the points where Y is smooth over S is an induced open subgroup of Y . By Exposé X, 3.5, the morphism $Y' \rightarrow X$ is an étale monomorphism and hence an open immersion. Since the fibres of X are connected and every open subgroup of an algebraic group is also closed, this open immersion is surjective, and therefore an isomorphism. Thus $Y' = X$, and a fortiori $Y = X$, completing the proof of 6.10.

Arguing as in Exposé VIII, 6.5, we deduce:

Corollary 6.11. — *Let G be an S -group prescheme locally of finite type and quasi-separated over S , let H be a group prescheme smooth and of finite presentation over S , with connected fibres, and let $i : H \rightarrow G$ be a monomorphism of S -groups, so that H is a subgroup of G . Then:*

- (a) $\text{Centr}_G(H)$ and $\text{Norm}_G(H)$ are representable by closed subpreschemes of G of finite presentation over G . Likewise, for every monomorphism $j : K \rightarrow G$ of finite presentation between S -group preschemes, $\text{Transp}_G(H, K)$ is representable by a closed subprescheme of G of finite presentation over G .
- (b) If G is quasi-compact, then in each of the cases in (a) there exists an integer $n \geq 0$ such that, if H_n denotes the order- n infinitesimal neighbourhood of the unit section of H (and K_n is defined similarly in the transporter case),

$$\begin{aligned}\text{Centr}_G(H) &= \text{Centr}_G(H_n), \\ \text{Norm}_G(H) &= \text{Norm}_G(H_n), \\ \text{Transp}_G(H, K) &= \text{Transp}_G(H_n, K) = \text{Transp}_G(H_n, K_n).\end{aligned}$$

Apply 6.10 over the base prescheme G to the group prescheme

$$X = H_G = H \times_S G.$$

For the centralizer, take as Y the inverse image of the diagonal subgroup of $(G \times_S G)_G$ under a suitable homomorphism of G -groups from X to $(G \times_S G)_G$. For the transporter, take the inverse image of K_G under a suitable homomorphism of G -groups from X to G_G . The normalizer case reduces to the transporter case by taking $K = H$. The hypothesis on G ensures that $H \rightarrow G$ is of finite presentation, and hence that $Y \rightarrow X$ is of finite presentation. In the centralizer case, the same hypothesis ensures that the diagonal subgroup of $G \times_S G$ is of finite presentation over $G \times_S G$; again it follows that Y is of finite presentation over X .

Remark 6.12. — One can prove, using rather delicate results of EGA VI,²⁸ that if X is a prescheme flat and of finite presentation over S , and is either proper over S or has nonempty connected geometric fibres, then for every closed subprescheme Y of X of finite presentation over S , the functor $\prod_{X/S}(Y/X)$ is representable by a closed subprescheme of S of finite presentation over S . Likewise, if X is an S -group prescheme flat and of finite presentation with connected fibres, and $i : Y \rightarrow X$ is a monomorphism of S -groups, with Y an S -group prescheme of finite presentation, then $\prod_{X/S}(Y/X)$ is representable by a closed subprescheme of S of finite presentation over S . In particular, 6.11(a) remains valid if the hypothesis “ H smooth over S ” is replaced by “ H flat over S .”

²⁸Cf. also J. P. Murre, “Representation of unramified functors. Applications,” *Séminaire Bourbaki*, no. 294 (May 1965), Theorem 3, p. 13.